



PHILIPPINE CAN SATELLITE
ROCKET COMPETITION



Design, Structural Evaluation, and Imaging Assessment of a Can Satellite for Biological Payload Stability Under Descent Deceleration and Vibration

Team Aetherpod



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ROCKET COMPETITION 2025



Table of Contents

CanSat Critical Design Review	4
1 Team Organization	4
2 Team Roles	5
3 Mission Overview.....	5
3.1 Scientific and Technical Objectives.....	5
3.2 System Requirements Summary	6
3.3 System Concepts of Operations	9
4 Acronyms.....	10
5 Sensor Subsystem Design.....	11
5.1 Air Pressure Sensor Summary	11
5.2 GPS Sensor Summary	12
5.3 Air Temperature Sensor Summary	14
5.4 Rotational Control Sensor Summary.....	15
5.5 CanSat Voltage Sensor Summary	15
5.6 Sound Vibration Summary.....	16
5.7 Camera Summary	16
6 Descent Control Design	17
6.1 Descent Rate of Parachute Prototype Testing	18
6.2 Descent Control Hardware Summary	18
6.3 Descent Rate Estimates	18
7 Mechanical Subsystem Overview	20
7.1 CanSat Main Frame Mechanical Layout	20
7.2 CanSat Cover Frame and Power Assembly Mechanical Layout	21
7.3 Payload and CanSat Electronics Mechanical Layout	21
7.4 CanSat Parachute Attachment Mechanism	22
7.5 Structure Survivability	22
7.6 Mass Budget	22
8 Communication and Data Handling Subsystem Design	24
8.1 CanSat Overview.....	24
8.2 CanSat Processor and Memory Selection	24
8.3 CanSat-to-Ground Radio Antenna.....	25
8.4 CanSat Telemetry Format.....	25
9 Electrical Power Subsystem Design.....	26
9.1 CanSat EPS Component	26
9.2 CanSat Circuit Diagram	27



9.3	CanSat EPS Diagram	28
9.4	CanSat Electrical Block Diagram	28
9.5	CanSat Power Source	29
9.6	CanSat Power Budget	29
10	Flight Software (FSW) Design	30
10.1	FSW Overview.....	30
10.2	CanSat FSW State Diagram	31
10.3	Software Development Plan	32
11	Ground Control System (GCS) Design	33
11.1	GCS Overview.....	33
11.2	GCS Antenna	33
11.3	Antenna Link Budget	34
12	CanSat Integration and Test.....	34
12.1	Subsystem Level Testing Plan.....	35
12.1	Integrated Level Functional Test Plan.....	38
12.2	Environmental Test.....	39
13	Test Procedures Descriptions.....	41
14	Mission Operation & Analysis	43
14.1	Mission Sequence of Events.....	43
14.2	Field Safety Rules Compliance	44
14.3	CanSat Location and Recovery	45
15	Requirement Compliance.....	45
16	Management.....	48
16.1	Mechanical Procurement	48
16.2	Electronics Procurement.....	48
16.3	Ground Station Procurement	48
16.4	CanSat Budget – Mechanics Components	49
16.5	CanSat Budget – Electrical & Electronics Components	49
16.6	CanSat Budget – Ground Station Components.....	50
16.7	CanSat Budget – Total Hardware Cost.....	50
16.8	CanSat Budget – Other Cost	50
16.9	CanSat Budget – Income.....	50
16.10	Detailed Program Schedule	51
17	Shipping and Transportation	51
18	Conclusion	51



CanSat Critical Design Review

The CDR should include evidence of prototypes through photos or CAD sketches. Think about including high level systems diagrams, circuit diagrams, block diagrams etc. Explain what sensors are being used and the justification for using them. Include block and flow diagrams and a high-level description of the design as the software is starting to be put together and tested. A full copy of the code is not required. Include parachute design, calculations used to reach the final decision and information on any prototypes developed.

1 Team Organization

Members	Information
 Jann Rovic Cueto	He is a 3 rd year Astronomy student major in Space Science Technology at Rizal Technological University, Philippines. Specializing in python programming, remote sensing application and basic satellite development.
 Roland Dayrit	He is a 4 th year Electronics Engineering student at Adamson University, specializing in electronic and parachute design of the CanSat.
 John Allen Tandoc	He is a 4 th year Electronics Engineering student at Adamson University, specializing in system design, etching and electrical wiring.
 Cyrill Cuna	He is a 2 nd year Computer Science student at Adamson University, specializing in Arduino programming of the sensors, MCU and basic wiring connection.
 Frank Fitzgerald Batin	He is a Technical Research Specialist at DOST-Adamson Star LAB and serves as the adviser for the team in completing the Can Satellite. He specializes in CAD, electronics, and satellite development.



2 Team Roles

Roles	Member Name
Mission Control Officer - Responsible for informing the Flight Coordinator when the team and their CanSat is ready to be launched.	1. Frank Fitzgerald Batin
Ground Station Crew - Responsible for monitoring the ground station for telemetry reception and issuing commands to the CanSat.	1. Jann Rovic Cueto 2. Cyrill John Cuña 3. Roland Dayrit
Recovery Crew - Responsible for tracking the CanSat and going out into the field for recovery and interacting with the field judges.	1. Jann Rovic Cueto 2. John Allen Tandoc
CanSat Crew - Responsible for preparing the CanSat, integrating it into the rocket, and verifying its status.	1. Roland Dayrit 2. John Allen Tandoc 3. Cyrill John Cuña

3 Mission Overview

3.1 Scientific and Technical Objectives

Mission Objectives
1. CanSat shall collect the required telemetry at a 1Hz sample rate and transmit the telemetry data to the ground station.
2. The CanSat shall be launched to an altitude ranging from 725m meters to 1 kilometer above the launch site and deployed near apogee.
3. After CanSat is deployed from the rocket, the CanSat shall descent using a parachute at a descent rate of 8 m/s.
4. During that time, the CanSat shall maintain the orientation of a video camera pointing in the south direction and the video camera shall be pointed 45 degrees downward to assure terrain is in the video.
5. Our team goal for this year is to be second place in Philippine Can Satellite Rocket Competition 2025.
6. To gain experience in working on engineering project, adapting to a teamwork environment, implementing project and time management.



7. The primary sensors shall gather the atmospheric status during ascent and descent of the CanSat.

Payload Objectives

1. To payload camera shall record the observations in the morphology change of the selected biological membrane.
2. The Gyroscope sensor shall record the triple axis orientation of the CanSat during ascent and descent phase.
3. The accelerometer shall record the ascent, descent acceleration and frequency vibration throughout the mission.

3.2 System Requirements Summary

RT	Requirements	Reasons	Priority	Verification			
				A	I	T	D
RT#1	All the components of the CanSat must fit a standard soft drink can (115 mm height and 66 mm diameter), with the exception of the parachute.	ESA Requirement	Medium		✓	✓	✓
RT#2	Radio antennas and GPS antennas can be mounted externally on the top or bottom of the can, depending on the design, but not on the sides.	ESA Requirement	Medium		✓	✓	✓
RT#3	The antennas, transducers and other elements of the CanSat cannot extend beyond the cans diameter until it has left the launch vehicle.	ESA Requirement	High		✓	✓	✓
RT#4	The mass of the CanSat must be between a minimum of 300 grams and a maximum of 350 grams +/- 10 grams. CanSat that are lighter must take additional ballast with them to reach the 300 grams minimum mass limit required.	ESA Requirement	High	✓	✓	✓	
RT#5	Explosives, detonators, pyrotechnics, and inflammable or dangerous materials are strictly forbidden. All materials used must be safe for the personnel, the equipment, and the environment.	ESA Requirement	High		✓		
RT#6	The CanSat must be powered by a battery and/or solar panels. It must be possible for the systems to remain switched on for four continuous hours.	ESA Requirement	High	✓	✓	✓	✓
RT#7	The battery must be easily accessible in case it has to be replaced/recharged.	ESA Requirement	Medium		✓	✓	



RT#8	The CanSat must have an easily accessible master power switch.	ESA Requirement	Medium		✓		✓
RT#9	Global Positioning System or GPS and a Beeper is required in the CanSat.	IAU Requirement	High		✓	✓	✓
RT#10	The CanSat should have a recovery system, such as a parachute, capable of being reused after launch. It is recommended to use bright colored fabric, which will facilitate recovery of the CanSat after landing.	ESA Requirement	High	✓			✓
RT#11	The parachute connection must be able to withstand up to 50 N of force. The strength of the parachute must be tested to ensure that the system will operate nominally.	ESA Requirement	High		✓	✓	✓
RT#12	A descent rate between 8 and 11 m/s is recommended for recovery reasons. However, the CanSat's descent speed must not be lower than 5 m/s or higher than 12 m/s for safety reasons.	ESA Requirement	High	✓		✓	
RT#13	The CanSat must be able to withstand an acceleration of up to 20g.	IAU Requirement	High		✓	✓	✓
RT#14	A Sd card module for data logger that saves data is required in the CanSat. And the cost of the CanSat must not exceed 8,000 pesos.	IAU Requirement	High	✓	✓		
RT#15	All electronics shall be hard mounted using proper mounts such as standoffs, screws, or high-performance adhesives.	Mission Requirement	High		✓	✓	
RT#16	All mechanisms shall be capable of maintaining their configuration or states under all forces.	Mission Requirement	High			✓	
RT#17	Both the container and payload shall be labeled with team contact information including email address.	Mission Requirement	High		✓		
RT#18	The container shall include a pressure sensor to measure altitude.	IAU Requirement	High	✓	✓	✓	
RT#19	The container shall measure its battery voltage.	Mission Requirement	High		✓	✓	
RT#20	The container shall transmit its telemetry once per second (1 Hz) in the formats described in the Telemetry Requirements section.	Mission Requirement	High			✓	
RT#21	The science payload shall include a video camera pointing directly in the microorganism sample.	Mission Requirement	High	✓	✓	✓	
RT#22	The CanSat shall include a video camera pointing 45 degrees up from the CanSat NADIR direction.	Mission Requirement	High	✓	✓	✓	
RT#23	The CanSat payload shall maintain orientation so the camera always faces south within +/- 20 degrees.	Mission Requirement	High	✓	✓	✓	



RT#24	The ground station shall command the CanSat to start transmitting telemetry prior to launch.	Mission Requirement	High		✓	✓	
RT#25	The ground station shall generate csv files of all sensor data as specified in the Telemetry Requirement Section.	Mission Requirement	High		✓	✓	✓
RT#26	Telemetry shall include mission time with one second or better resolution. Mission time shall be maintained in the event of processor reset during the launch and mission.	Mission Requirement	High	✓	✓	✓	✓
RT#27	Configuration states such as if commanded to transmit telemetry shall be maintained in the event of a processor reset during launch and mission.	Mission Requirement	High		✓	✓	
RT#28	Each team shall develop their own ground station.	IAU Requirement	High	✓	✓	✓	
RT#29	All telemetry shall be displayed in real time during descent on the ground station.	IAU Requirement	High	✓	✓	✓	✓
RT#30	All telemetry shall be displayed in engineering units (meters, meters/sec, Celsius, etc.)	IAU Requirement	High	✓	✓	✓	
RT#31	Teams shall plot each telemetry data field in real time during flight (optional).	IAU Requirement	High		✓	✓	
RT#32	The ground station shall include one laptop computer with a minimum of two hours of battery operation, HC12 radio and an antenna.	IAU Requirement	High		✓	✓	
RT#33	All video cameras shall be in color, have a resolution of at least 640x480 and record at a minimum of 30 frames a second.	Mission Requirement	High	✓			✓
RT#34	The ground station must be portable so the team can be positioned at the ground station operation site along the flight line. AC power will not be available at the ground station operation site.	Mission Requirement	High		✓		
RT#35	The CanSat must operate during the environmental tests.	IAU Requirement	High	✓	✓	✓	✓
RT#36	The flight software shall maintain a count packet transmitted which shall increment with each packet transmission throughout the mission. The value shall be maintained through processor resets.	Mission Requirement	High	✓		✓	✓
RT#37	The CanSat shall maintain mission time throughout the whole mission even with processor resets or momentary power loss.	Mission Requirement	High	✓		✓	✓



RT#38	The container shall have its time set to UTC time to within one second.	Mission Requirement	High		✓	✓	
RT#39	The flight software shall use the radio uplink pressure values in place of the pressure sensor for determining the CanSat altitude.	Mission Requirement	High			✓	✓

3.3 System Concepts of Operations

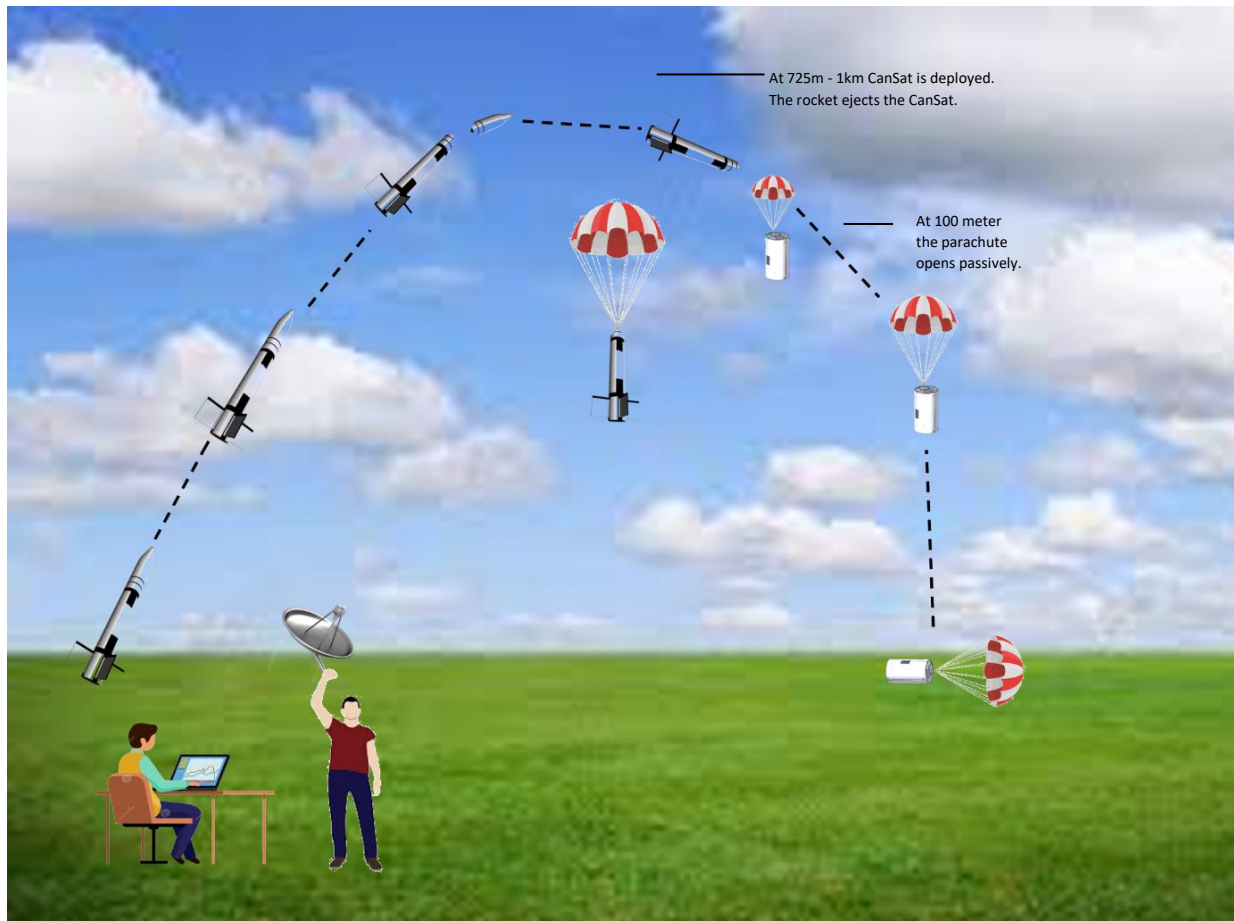
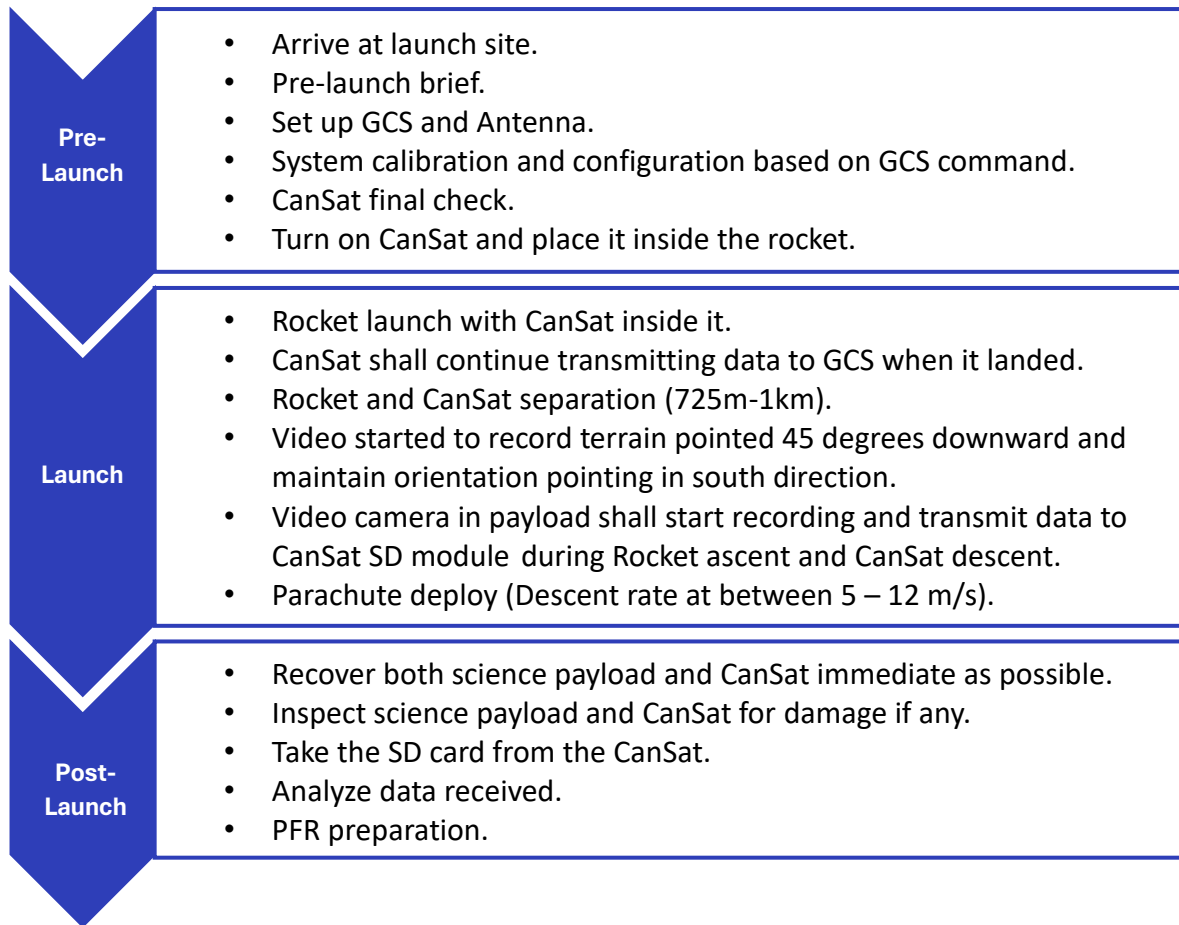


Figure 1. Visualization of concept operations of our CanSat.

Our Can Satellite (CanSat) mission follows a structured sequence of operations to achieve its objectives. The CanSat is deployed from a predetermined altitude using a rocket launch vehicle, initiating its descent under a controlled parachute system. During descent, onboard sensors continuously collect and transmit environmental and flight data, including temperature, pressure, altitude, and acceleration, to the ground station. The structural design ensures payload stability and protection against descent forces and vibrations. Data acquisition and transmission processes are monitored in real time to assess system performance and mission success. The collected data is then analyzed to evaluate the CanSat's operational efficiency and payload integrity.



4 Acronyms

3D	Three Dimensional	GPS	Global Positioning System
AC	Alternating Current	I/O	Input to Output
CAD	Computer Aided Design	I2C	Inter-Integrated Circuit
CDH	Communication Data Handling	I&T	Integration & Test
CDR	Critical Design Review	IDE	Integrated Development Environment
CSV	Comma Separated Value	IMU	Inertial Measurement Unit
dB	Decibel	Li-Ion	Lithium-Ion
dBi	Decibel Isotropic	MCU	Microcontroller Unit
DC	Direct Current	ME	Mechanical Subsystem
DCS	Descent Control System	PCB	Printed Circuit Board
EPS	Electrical Power System	RAM	Random Access Memory
FSW	Flight Software	RN	Requirement Number
G	Gravitational Force	SS	Sensor Subsystem
GCS	Ground Control Station	SPI	Serial Peripheral Interface
GPIO	General Peripheral Input/Output	UART	Universal Asynchronous Receiver/Transmitter (Serial)
GND	Ground	VM	Verification Method



A	Analysis	D	Demonstration
I	Inspection	T	Test

5 Sensor Subsystem Design

Sensor Type	Model	Function	Located In
Air Pressure	MPL3115A2	To measure air pressure, the collected data will be converted to altitude	CanSat
Air Temperature	MPL3115A2	To measure temperature inside of container and payload	CanSat
GPS	UBLOX – Neo M8N	To acquire position or location	CanSat
Three Axis Sensors	MPU6050	To measure gyro and accelerometer	CanSat
Camera	ESP32 CAM	To record video	Payload & CanSat
Microphone Amplifier	MAX4466	To record vibration sounds	CanSat

5.1 Air Pressure Sensor Summary

Model	Range (kpa)	Sensitivity (kpa)	Voltage (V)	Current (μA)	Interface	Mass (g)	Accuracy (hpa)	Size (mm)	Cost (₱)
MPL3115A2	20-110	0.0625	1.62 – 3.6	8.5	I2C	1.2	± 0.4	18x19x2	645

Data Format	(Altitude, m)
Equations	$Altitude = 44330 \times \left(1 - \left(\frac{P}{P_0}\right)^{\frac{1}{5.255}}\right)$ <i>P: Atmospheric pressure at current altitude (in hPa)</i> <i>P₀: Atmospheric pressure at sea level (in hPa)</i>
Sample Output	<pre>Altitude = 121.9m Altitude = 122.2m Altitude = 122.0m</pre>

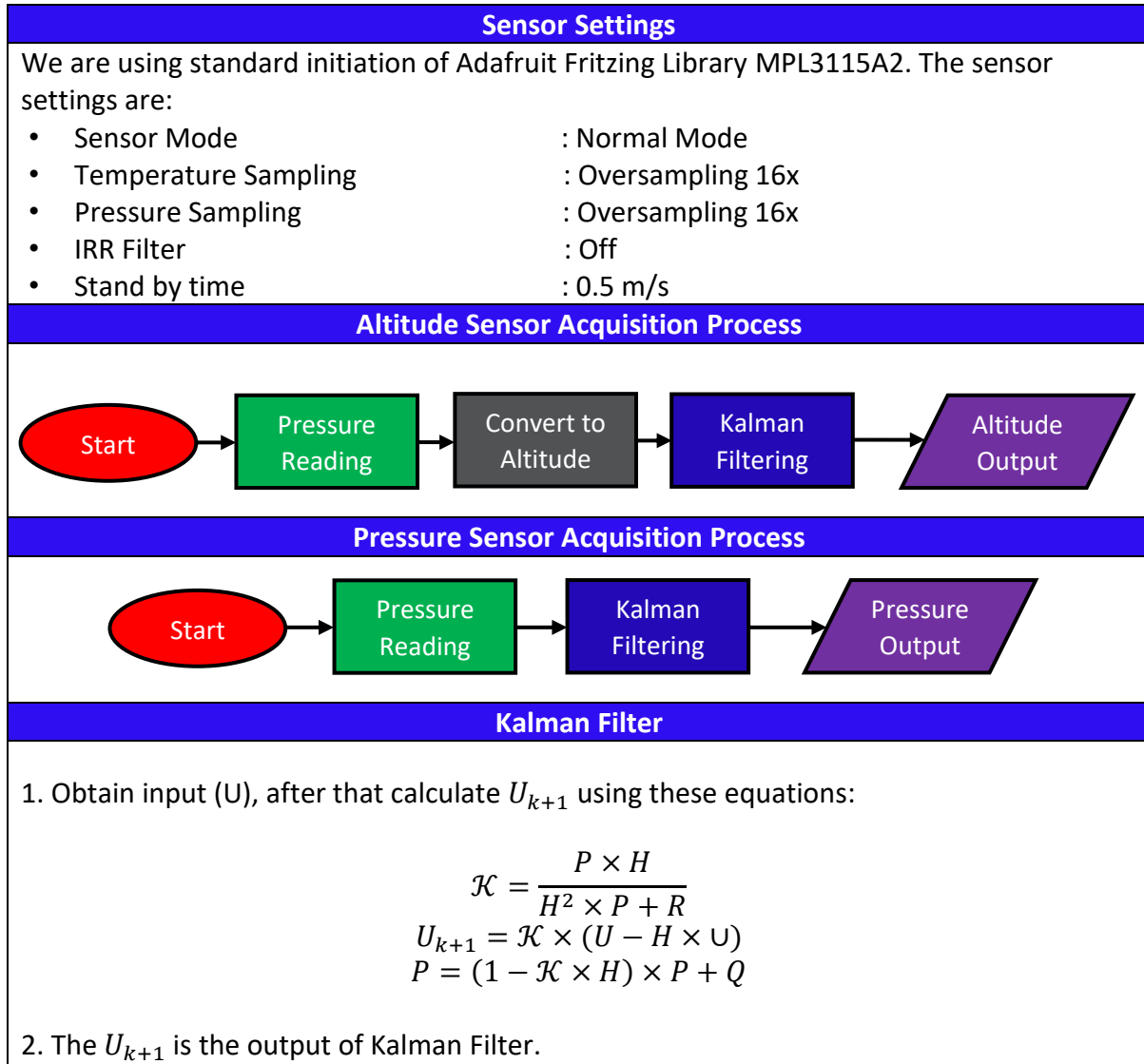


MPL3115A2 will collect pressure data according to the settings that have been set. After that the pressure data will be converted into altitude data.



Sample packet output. See example below:

19:08:41:300| 120.984964, 14.584964|-9.09,-0.22,-1.96|-0.04,0.02,0.02|49.75|30.49|54.64|1534|1013.25|122.9



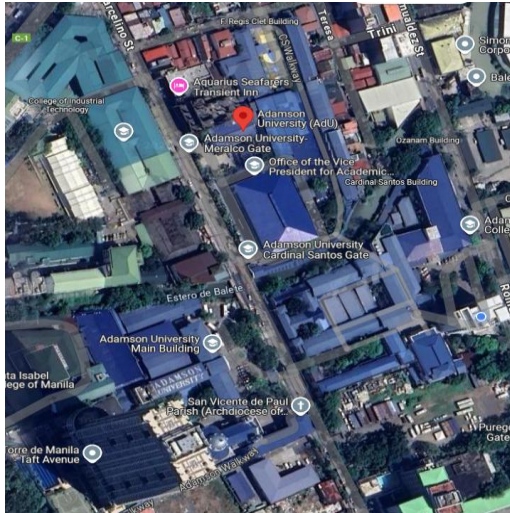
5.2 GPS Sensor Summary

Model	Range (m)	Sensitivity (dBm)	Voltage (V)	Current (mA)	Interface	Mass (g)	Accuracy (m)	Size (mm)	Cost (₱)
UBLO XNEO-M8N	As long can connect to satellites	-162	2.7 - 3.6	67	UART	16.8	± 2.5	16x 12.2 x2.4	600



Data Format	(Time, hh: mm: ss) (Latitude, °) (Longitude, °) (Altitude, m) (Satellites, integer value showing number of satellites what was connected)
Equations	It will automatically calculate the distance of this sensor with reference to satellite (at least 3)
Sample Output	<pre>GPS TIME:19:50:17 GPS LATITUDE :-7.2936 GPS LONGITUDE :112.8079 GPS ALTITUDE :19.9 GPS SATELITES :6</pre>



Sensor Settings	
We are using a standard initiation of TinyGPS++ library. The sensor settings is already define by manufacturer, the default mode are: - Baud Rate : 9600 baud - Data : 8 bits - Stop : 1 stop bit - Parity : None - Output Protocol : NMEA	
GPS Test	
	Longitude reading: 120° 59' 5.99" E Latitude reading: 14° 35' 5.99" N Longitude original: 120° 59' 5.99" E Latitude original: 14° 35' 5.99" N

Sample packet output. See example below:

19:08:41:300|120.984964, 14.584964|-9.09,-0.22,-1.96|-0.04,0.02,0.02|49.75|30.49|54.64|1534|1013.25|122.9



5.3 Air Temperature Sensor Summary

Model	Range (°C)	Sensitivity (°C)	Voltage (V)	Current (μA)	Interface	Mass (g)	Accuracy (°C)	Size (mm)	Cost (₱)
MPL3115A2	(-40) – (+85)	0.01	1.62 – 3.6	8.5	I2C	1.2	± 1	18x19x2	645

Data Format	(Temperature, °C)
Equations	There is no equation for this measurement.
Sample Output	<pre>Temperature = 28.5°C Temperature = 28.5°C Temperature = 28.5°C</pre>



MPL3115A2 will collect pressure data according to the settings that have been set. After that the pressure data will be converted into altitude data.

Sample packet output. See example below:

19:08:41:300|120.984964, 14.584964|-9.09,-0.22,-1.96|-0.04,0.02,0.02|49.75|30.49|54.64|1534|1013.25|122.9

Sensor Settings

We are using standard initiation of Adafruit Fritzing Library MPL3115A2. The sensor settings are:

- Sensor Mode : Normal Mode
- Temperature Sampling : Oversampling 16x
- Pressure Sampling : Oversampling 16x
- IRR Filter : Off
- Stand by time : 0.5 m/s

Temperature Sensor Acquisition Process

```

graph LR
    Start([Start]) --> Reading[Temperature Reading]
    Reading --> Average[Moving Average]
    Average --> Output[/Temperature Output/]

```

Moving Average

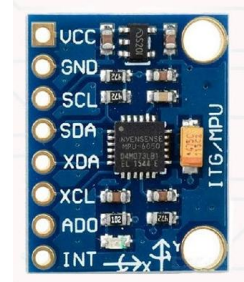
1. Obtain first 3 data of input (k=3), after that calculate the output using this equation:
$$MA = \frac{1}{k} \sum_{i=n+k-1}^k P_i$$
2. After that we must throw the oldest data and replace it with newer data and perform next calculations.
3. Do it continuously.



5.4 Rotational Control Sensor Summary

Model	Sensitivity	Voltage (V)	Current (μA)	Interface	Mass (g)	IMU	Size (mm)	Cost (₱)
MPU6050	12.3	2.3 – 3.4	12.3	I2C	2	3-axis	5.2x3.8x1.1	85

Range	Sensitivity
Gyro (2000dps) Accelerometer (16G)	Gyro (16bits) Accelerometer (14bits)



Data Format	(Gyrometer Roll, °), (Gyrometer Pitch, °), (Gyrometer Yaw, °) (Accelerometer Roll, °), (Accelerometer Pitch, °), (Accelerometer Yaw, °)
Equations	Pointing Errors = Absolute Orientation – South Orientation
Sample Output	Gyro: x= 0.002 y= 0.008 z= -0.038 Accel: x= 9.320 y= -1.700 z= -2.470

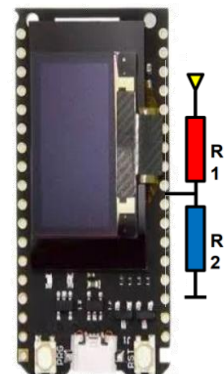
Sample packet output. See example below:

19:08:41:300|120.984964, 14.584964|-9.09,-0.22,-1.96|-0.04,0.02,0.02|49.75|30.49|54.64|1534|1013.25|122.9

5.5 CanSat Voltage Sensor Summary

Model	Range (V)	Sensitivity (V)	Voltage (V)	Current (μA)	Interface	Mass (g)	Accuracy (%)	Size (mm)	Cost (₱)
ESP32 OLED V2.0	0-3.3	0.000805	Embedded	Embedded	I2C	Embedded	± 2	Embedded	1,099

Data Format	(Battery Voltage, V)
Equations	$\text{Battery Voltage} = \frac{3.3}{1024} \times \frac{R_1 + R_2}{R_2} \times \text{ADC Read}$ R_1: First Resistor R_2: Second Resistor





Sample Output	Battery Voltage = 3.29V Battery Voltage = 3.29V Battery Voltage = 3.28V
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Sensor Settings
From our calculation, we choose 1000 ohms as our R_1 and 3300 ohms as our R_2, the following is the detail of our calculation: $V_{\max batt}: V_{adc}$ $4.2 \text{ V}: 3.3 \text{ V}$ $R_1 = 4.2 \text{ V} - 3.3 \text{ V} = 0.8 \times 1000 = 800\Omega \approx 1000\Omega$ $R_2 = 3.3 \text{ V} \times 1000 = 3300\Omega$
Voltage Sensor Acquisition Process
<pre> graph LR Start([Start]) --> Reading[Battery Voltage Reading] Reading --> Average[Moving Average] Average --> Output[/Battery Voltage Output/] </pre>
Moving Average
- Obtain first 3 data of input ($k=3$), after that calculate the output using this equation: $MA = \frac{1}{k} \sum_{i=n+k-1}^k P_i$ - After that we must throw the oldest data and replace it with newer data and perform next calculations. - Do it continuously.

5.6 Sound Vibration Summary

Model	Gain (kHz)	Voltage (V)	Current (μA)	Sensitivity (dB)	Mass (g)	Size (mm)	Cost (₱)
MAX4466	200-600	2.4 – 5.5	24	-44 $\pm$ 2	0.80	9.7 x 4.5	529



MAX4469 are micropower op amps optimized for use as microphone preamplifiers. They provide the ideal combination of an optimized gain bandwidth product vs. supply current, and low voltage operation in ultra-small packages.

5.7 Camera Summary

Model	Resolution (pixels)	Frames per second	Voltage (V)	Current (mA)	Interface	Mass (g)	Field of View (°)	Size (mm)	Cost (₱)
ESP32-CAM	800 x 600	30	3.3 – 5.0	140	UART	20	65	40.5x27 x4.5	654

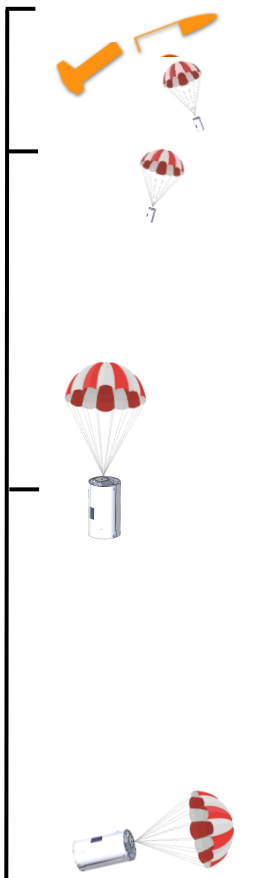
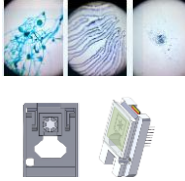


Output Camera	
The camera will be turned on and start recording the video, on the launch pad, to record during both the ascent and landing of the payload. It will capture the horizontal view in one direction. And the other camera will record the payload status of microorganisms. The video is stored on a microSD card.	
	



ESP-32 CAM

6 Descent Control Design

	Container separates from the rocket at 725-1000 m. The container will descent 8m/s (± 5 m/s) using the main parachute.	 CanSat	Material: PETG thermoplastic polyester Dimension: 115 mm in height and 66 mm in diameter. Feature: Parachute attachment, the 3d printed holder of payload can slid the glass cover and glass slides.
	The CanSat maintain the orientation of a video camera pointed 45 degrees in the south direction.	 Parachute	Material: Black/Gray ripstop nylon. Dimension: 129 mm in diameter. Feature: Spill hole as stabilizer with 10.32 mm in diameter.
	The container landed and telemetry transmission will be stopped.	 Payload	Material: Glass slide, glass cover. Dimension: 25mmx75mm Feature: Plain and can hold a wide range of specimens.



6.1 Descent Rate of Parachute Prototype Testing

CanSat Main Parachute Descent		Documentations
Goal	The CanSat shall descend using a parachute. The parachute must be at a rate of between 5 m/s – 12 m/s.	
Procedure	1. The parachute was tied to the dummy CanSat with a weight of 340g). 2. The object was dropped 50m from the drone. 3. Observe and measure time elapsed for the object land.	
Result	It takes time to be fully stable descend with a descent rate of ± 8 m/s using the parachute.	

6.2 Descent Control Hardware Summary

	Key Design Considerations: • The CanSat focuses mass at the bottom. • The type of parachutes is round parachute with a spill hole. Color Selection: • Parachutes are using black nylon. • The cover of the CanSat is yellow PETG composite. Passive Component: • The parachute is attached to CanSat directly to top of the middle frame.
---	---

6.3 Descent Rate Estimates



The descent rate of descent phase will be estimated using different parameters

Main Parachute

Parameters: - Diameter of Parachute (D_p)

- Diameter of Spill Hole (D_{sh})

- Descent rate of 7 m/s – 12 m/s ($\pm$ m/s)

We use the range of descent velocity between minimum [$V = 5\text{m/s}$] and maximum [$V_{\text{max}} = 12\text{m/s}$] to determine diameter of parachute.

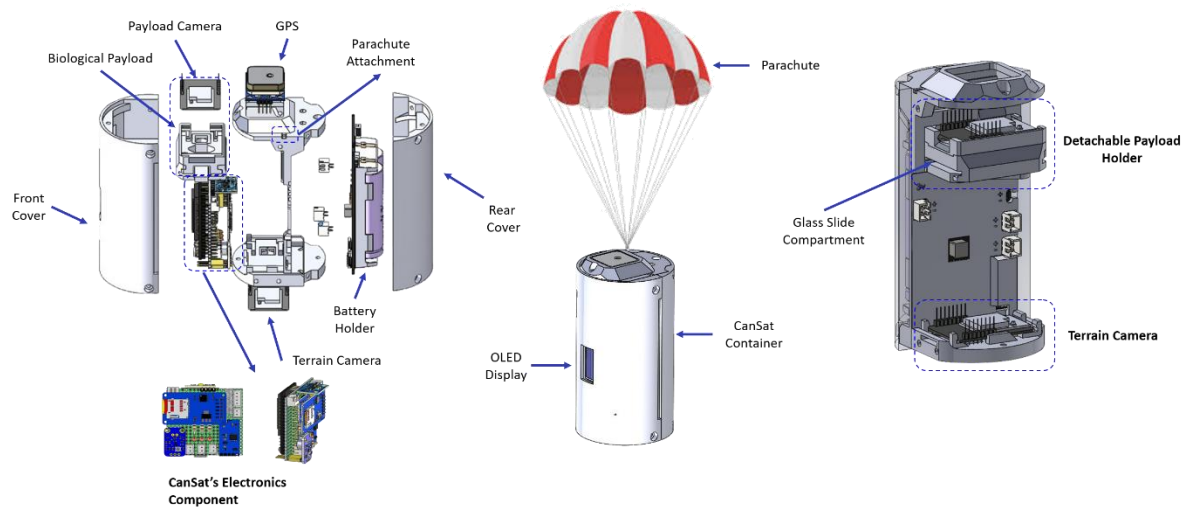


Minimum Velocity Speed [m/s]	Maximum Velocity Speed [m/s]
$v = \sqrt{\frac{8 \times m \times g}{\rho \times Dp^2 \times \pi \times C_d}}$ $v = \sqrt{\frac{8 \times (0.3) \times (9.8)}{1.225 \times (0.129) \times (3.14) \times (0.8)}}$ <i>v = 7.604 m/s</i> Information: Dp = The diameter of the parachute (m) v = Descent speed (m/s) π = 3.14 g = 9.8 m/s² Dsh = Spill hole (m) *Assumption: Cd = 0.8 (Coefficient drag of parachute) m = 0.3 kg (CanSat) ρ = 1.225 g/m³	$v = \sqrt{\frac{8 \times m \times g}{\rho \times Dp^2 \times \pi \times C_d}}$ $v = \sqrt{\frac{8 \times (0.35) \times (9.8)}{1.225 \times (0.129) \times (3.14) \times (0.8)}}$ <i>v = 8.214m/s</i> Information: Dp = The diameter of the parachute (m) v = Descent speed (m/s) π = 3.14 g = 9.8 m/s² Dsh = Spill hole (m) *Assumption: Cd = 0.8 (Coefficient drag of parachute) m = 0.3 kg (CanSat) ρ = 1.225 g/m³
Diameter of the spill hole is 8%-10% of parachute's diameter because we set tolerance of ±2%. Spill hole diameter = Dsh = Dp × 8% = 0.01032 m Spill hole radius = Dsh/2 = 0.00516 m	

Parachute Summary:	
Altitude	The descent rate descent phase will be estimated using parameters
1000 - 0 m	Type parachute: Round parachute The diameter parachute: 129 mm Spill hole diameter: 10.32 mm The descent speed: 8.214 m/s ≈ 8 m/s
CanSat Summary	
Using passive control for CanSat stability. Focus of mass in the bottom of the CanSat to keep the stability and decrease swing. It also helps maintain nadir direction.	

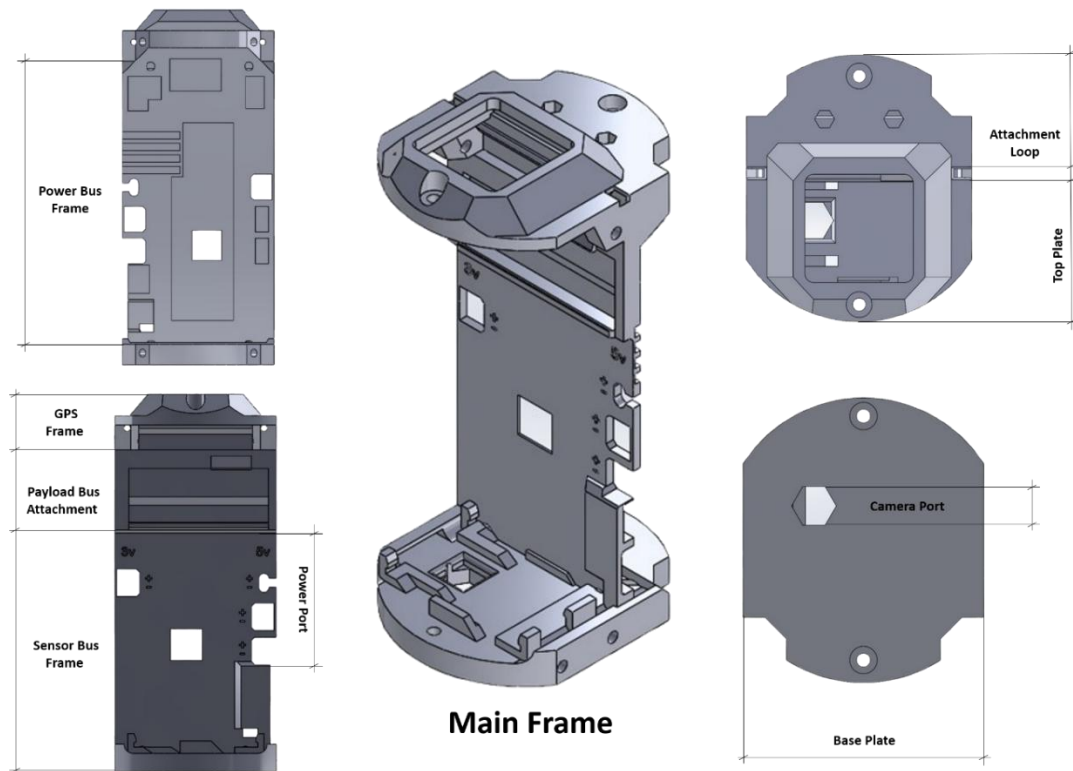


7 Mechanical Subsystem Overview



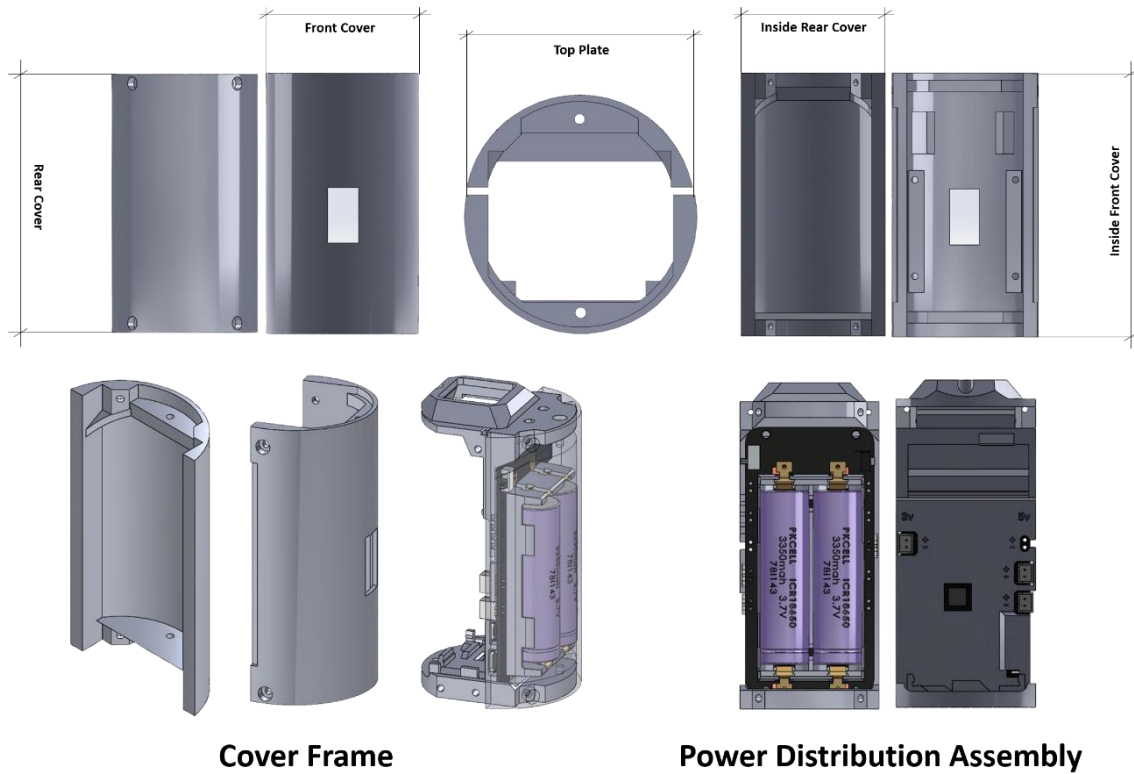
CanSat Container	CanSat Payload
Material: PETG Parachute: Black Nylon Note: The CanSat's top part of the main frame has a small hole for parachute attachment.	Material: PETG and Acrylic Note: The payload holder is detached from the upper part of the main frame, and the acrylic biological glass slide is securely fitted inside.

7.1 CanSat Main Frame Mechanical Layout

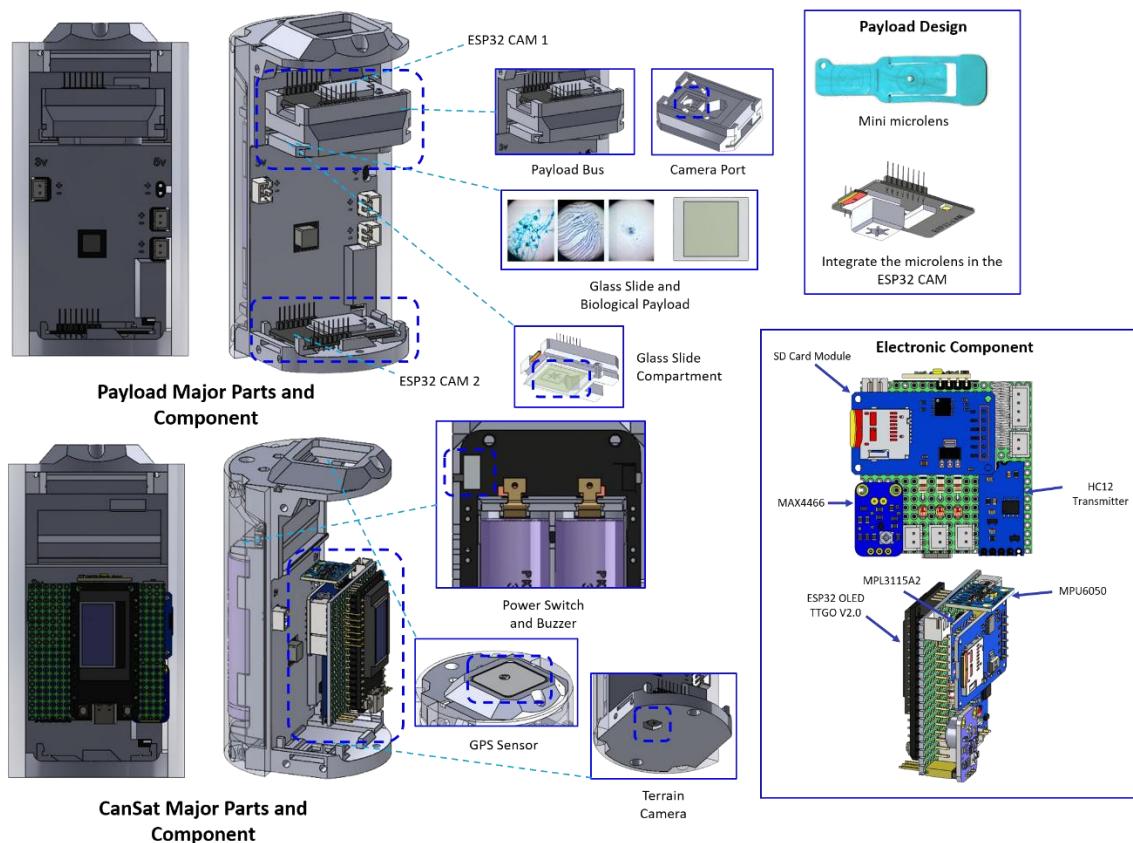




7.2 CanSat Cover Frame and Power Assembly Mechanical Layout

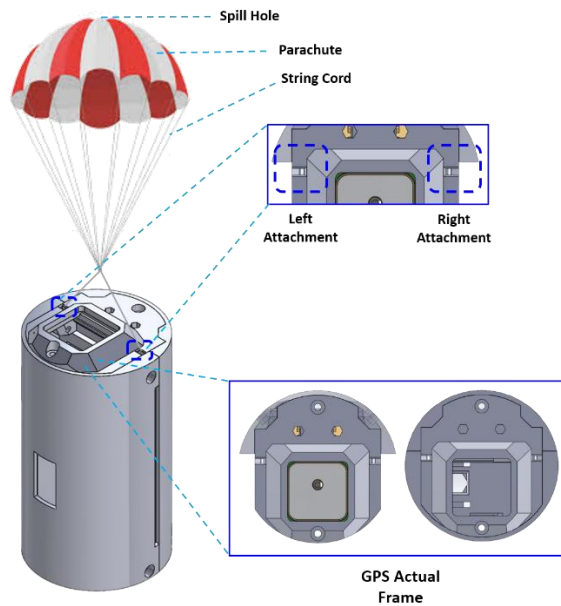


7.3 Payload and CanSat Electronics Mechanical Layout





7.4 CanSat Parachute Attachment Mechanism



Information

1. The parachute is placed on the Top of the CanSat at folded position.
2. The cord of parachute is carefully knot.
3. The parachute will be opened by the air-resistance assist.

7.5 Structure Survivability

Acceleration and Shock Force Requirement Testing	
Requirement	Documentation
CanSat survived given acceleration and shock free force from 61 cm feet attached cord. No electrical mounts disconnected, no parts falling off and no separation.	
1. Attach 61cm cord to the parachute. 2. Add a floor mat or pillow below the test fixture. 3. Drop the CanSat. 4. Observe if any parts of the CanSat fall apart.	
Result	
The CanSat did not lose power. Inspect for any damage, or detached parts and verify telemetry is still being received.	

7.6 Mass Budget

Electrical Component				
Component	Quantity	Mass Unit (g)	Mass (g)	Determination
UBLOX NEO M8N	1	17.2	17.2	Measured
Buzzer	1	1.7	1.7	Measured
PCB	1	13.9	13.9	Estimated
ESP32 OLED V2.0 TTGO	1	8.8	8.8	Measured



ESP32 -CAM	2	6.2	12.4	Measured
HC-12 Transmitter Module	1	2.2	2.2	Measured
MPL3115A2	1	1.6	1.6	Measured
Thermistor/Resistor	2, 3	0.01	0.01	Estimated
Battery	2	47.45	94.9	Measured
Wire Connectors	1	9.5	9.5	Measured
MPU6050	1	1.6	1.6	Measured
MAX4466 Amplifier	1	2.0	2.0	Measured
SD Card	3	0.25	0.75	Measured
Mobile Battery Holder	1	38.1	38.1	Measured
SD Card Module	1	3.9	3.9	Measured
Total Mass Electrical Components of CanSat			207.96	

Structural Component				
Component	Quantity	Mass Unit (g)	Mass (g)	Determination
Parachute	1	4.5	4.5	Estimated
Fish String	1	1	1	Estimated
Hex Screws	20	0.65	13	Measured
Specimen Chamber	1	15.1	15.1	Measured
Front frame	1	33.8	33.8	Measured
Middle Frame	1	33.3	33.3	Measured
Rear Frame	1	32.5	32.5	Measured
Total Mass Structural Component of CanSat			133.2	

Payload Component				
Component	Quantity	Mass Unit (g)	Mass (g)	Determination
Biological Membrane	1	0.05	0.05	Estimated
Macro Lens	1	0.05	0.05	Estimated
Glass Cover	1	2.2	2.2	Measured
Glass Slide	1	4.5	4.5	Measured
Total Mass Payload Component			6.8	

RN#4 Total mass of the CanSat (science payload and CanSat) shall be 300-350 grams ± 10 grams.

Total Mass	
CanSat	341.16
Payload	6.8
Total Mass of All System	347.96

Correction Method (Margin Competition ± 10)	
If Total Mass system < 340 gram	We will increase the mass of materials with higher infill density with 3D printed material for container or payload.

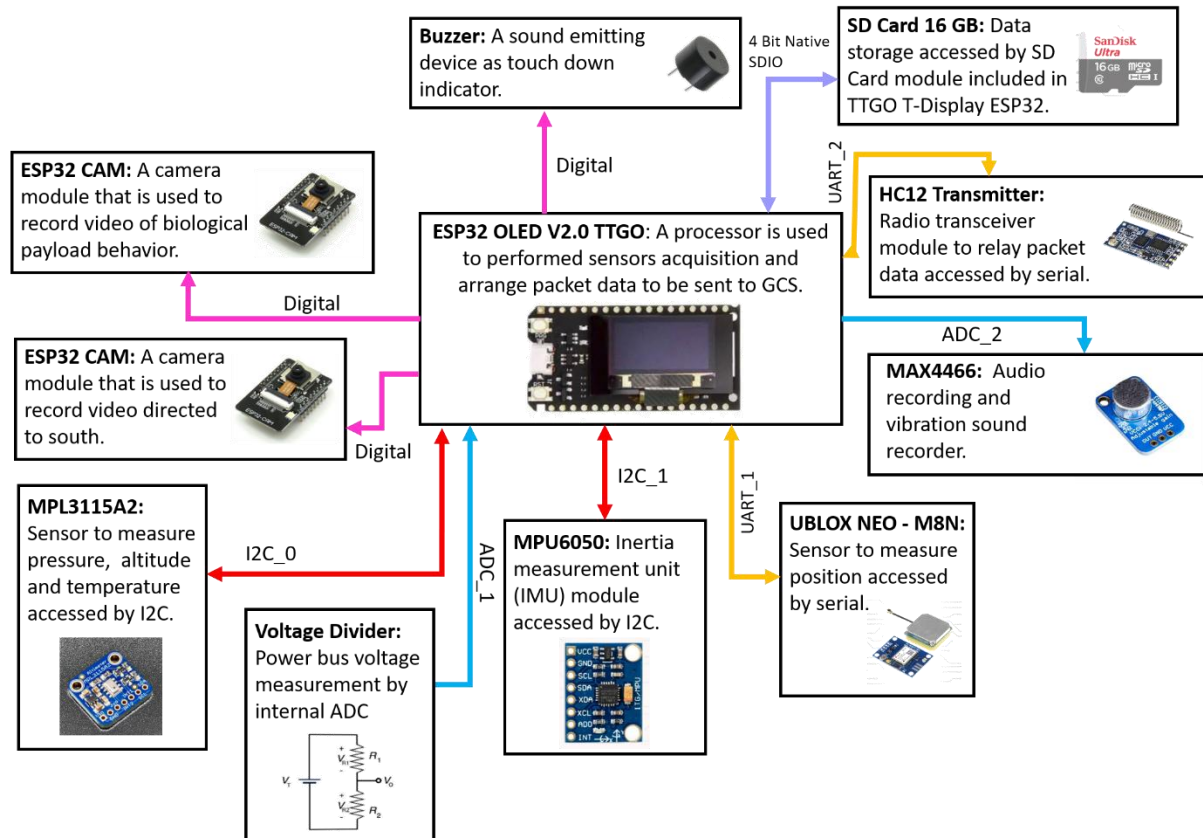


If Total Mass system > 350 gram

We will change the material with lighter material such as composite that has lower density for container and payload.

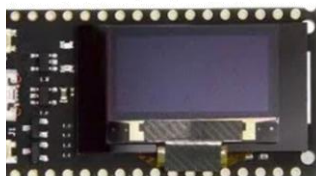
8 Communication and Data Handling Subsystem Design

8.1 CanSat Overview



8.2 CanSat Processor and Memory Selection

Model	Data bus width (bits)	Processor Speed (Mhz)	Power Consumption (mW)	Operating Voltage (V)	Size (mm)	Mass (g)
ESP32 OLED V2.0	32	600	500	3.6-5	50x25.4	8.8



Memory Storage	Boot Time (s)	Cost (₱)
EEPROM 4KB Flash 4MB RAM 820KB	0.005	1,099



Reasons
1. It can be programmed by Arduino IDE; this will make the software development process faster. 2. It has suitable size of Flash Memory to store longer programming code. 3. It has a window screen monitoring for fast checking of programmed results. 4. It has a processor with a high speed without sacrificing aspects such as temperature and power requirements.

Model	Memory (GB)	Interface	Data Transfer Rate		Cost (₱)
			Write (MB/s)	Read (MB/s)	
SanDisk Ultra	16	SD Card Interface	98	98	249



Reasons
1. More stable data transfer rate 2. It is more reliable and easier to use than using EEPROM. 3. System processor already has a peripheral to access it.

8.3 CanSat-to-Ground Radio Antenna

Model	Sensitivity (dBm)	Range (m)	Transmit Current (mA)	Supply Voltage (V)	Frequency (MHz)	Mass (g)	Size (mm)	Cost (₱)
HC12	-111	600-1000	100	3.2 – 5.5	433.4-473.0	2.2	27.4x13.2x4	295



Reasons
1. It has wider range compared to the other HC12 model. 2. It is more sensitivity compared to the other HC12 model.

8.4 CanSat Telemetry Format

CanSat Telemetry Data		
Data Format	Sample Data	Description
<CHANNEL_ID>	015	The assigned team channel.



<ALTITUDE>	100.1	The altitude in units of meters and must be relative to ground level. The resolution must 0.1 meters.
<PTEMP> <STEMP>	28.5 26.5	The temperature in degrees Celsius with a resolution of 0.1 degrees C.
<GPS_TIME>	00:09:03	The time generated by the GPS receiver. The time must be reported in UTC and have a resolution of second.
<GPS_LATITUDE>	17.5849	The latitude generated by the GPS receiver in decimal degrees with a resolution of 0.0001 degrees North
<GPS_LONGITUDE>	120.9849	The latitude generated by the GPS receiver in decimal degrees with a resolution of 0.0001 degrees West.
<GPS_ALTITUDE>	100.2	The altitude generated by the GPS receiver in meters above mean sea level with a resolution of 0.1 meters.
<GPS_SATS>	3	The number of GPS satellites being tracked by the GPS receiver. This must be an integer number.
<GYRO_R, GYRO_P, GYRO_Y>	1.120 2.120 3.120	The CanSat gyro readings in degrees per second for the roll, pitch, and yaw axes.
<ACCEL_R, ACCEL_P, ACCEL_Y>	1.120, 2.120, 3.120	The CanSat accelerometer readings for the roll, pitch, and yaw axes.
<SD_CARD_STATUS>	1	The SD Card status reading from 0 to 1.

Payload Data Format will be transmitted @1 Hz to the ground station as follows:
(015|100.1|28.5|00:09:03|17.5849,120.9849|100.2|3|1.120,2.120,3.120|1.120,2.120,3.120|1|

9 Electrical Power Subsystem Design

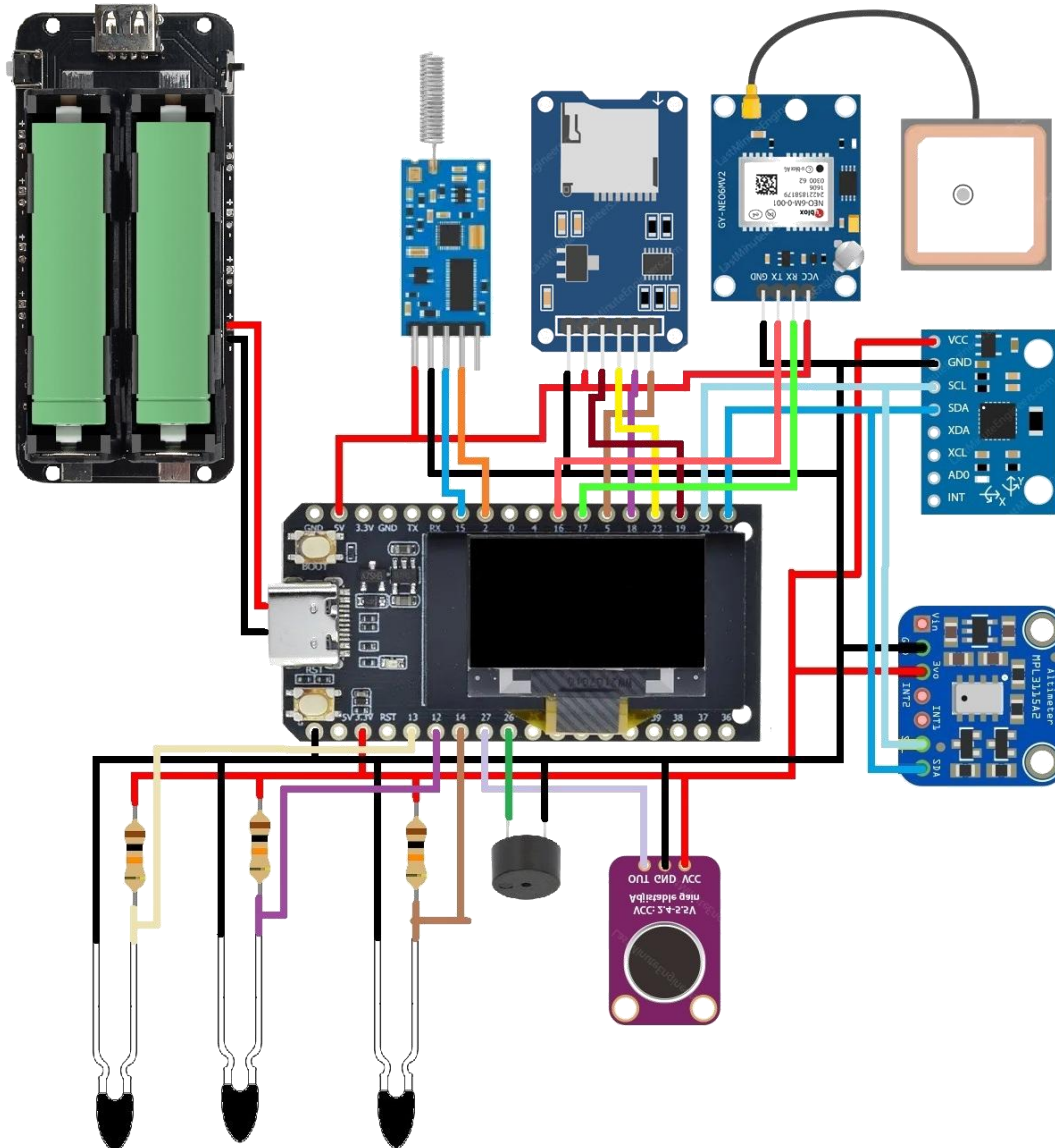
9.1 CanSat EPS Component

Component	Purpose
Power	1. An PKCell Lithium-ion 18650 3.7V battery is used to supply all components in the CanSat. 2. A power switch (external on/off switch) will be used to connect or disconnect the power and also used to reset the CanSat. 3. The 3.3V regulator is used to supply all sensors and HC-12 in the CanSat.
MCU	1. ESP32 OLED V2.0 TTGO will collect all data sensors. It will be supplied by 3.6V output.



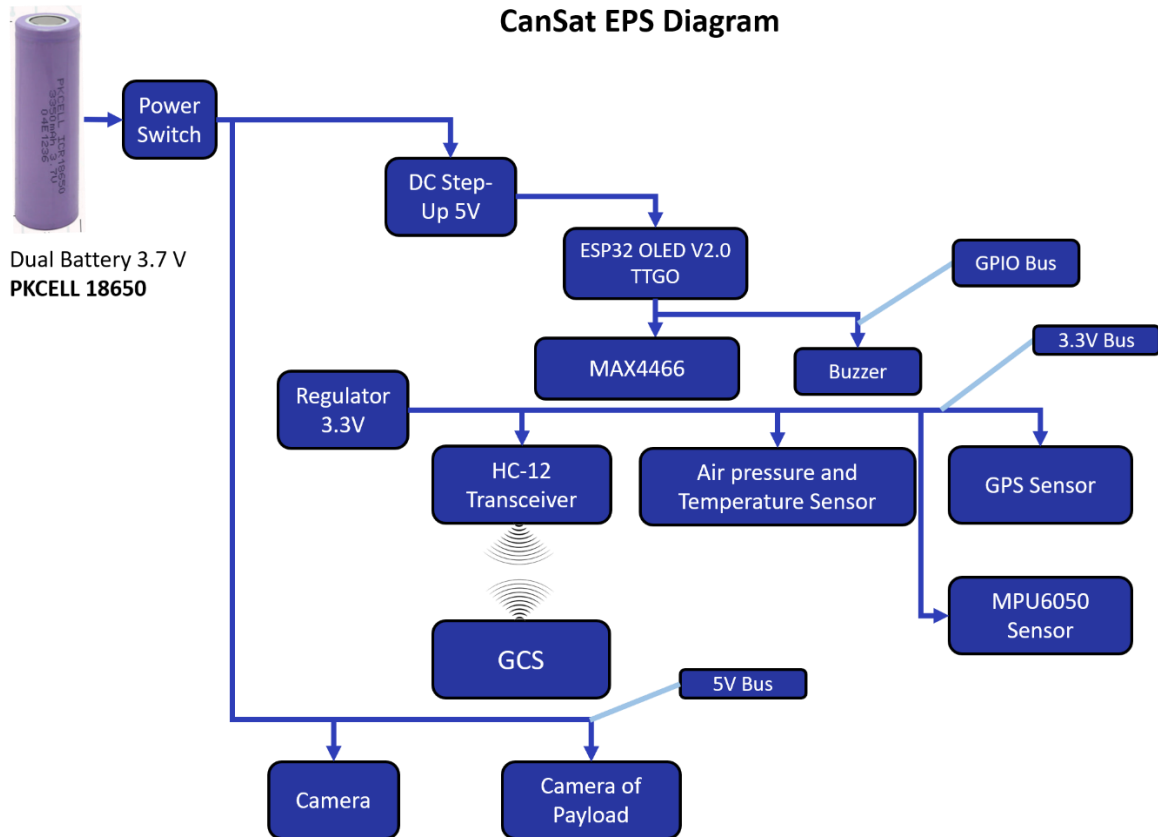
	2. SD Card will save all sensors data.
Sensors	1. MPL3115A2 will collect temperature and pressure data. It will be supplied by 3.3V from the battery power source. 2. Ublox NEO-M8N will acquire position data. It will be supplied by 3.3V from the battery power source. 4. The camera shall record the observation in organism. It will be supplied by 5 V directly from battery. 5. The buzzer will be on as a beacon when CanSat lands. It will be supplied by 3.3V from GPIO pin.
Communications	1. HC-12 will transmit and receive data from the GCS. It will be supplied by 3.3V from the battery power source.

9.2 CanSat Circuit Diagram

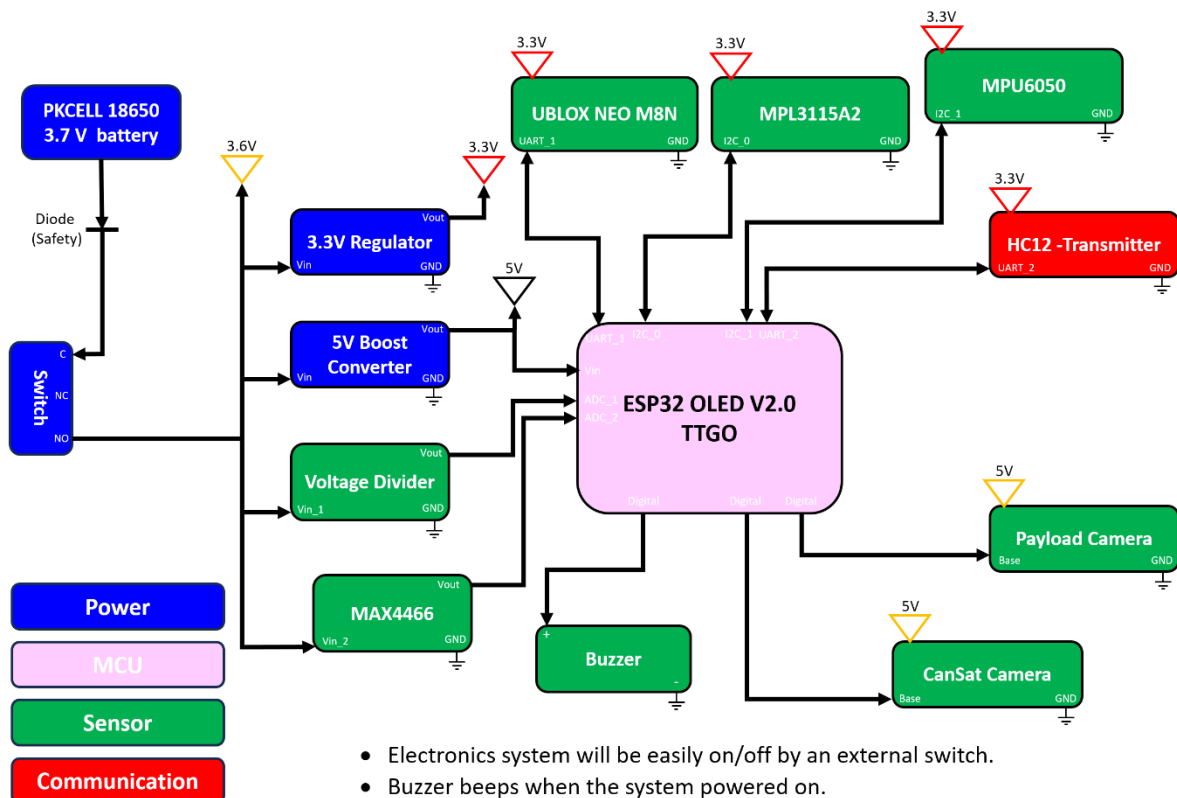




9.3 CanSat EPS Diagram



9.4 CanSat Electrical Block Diagram





9.5 CanSat Power Source

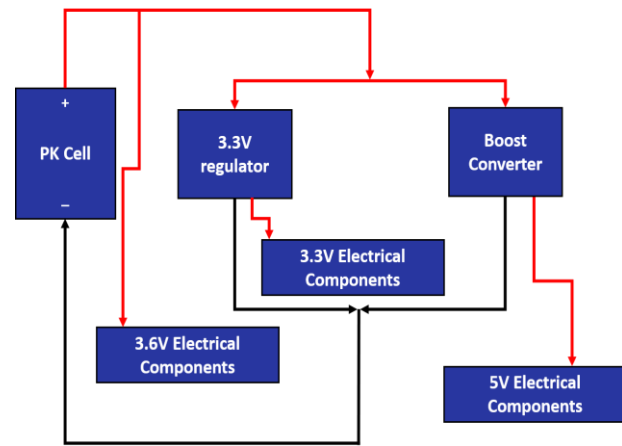
Model	Battery Chemistry	Voltage (V)	Nominal Capacity (mAh)	Maximum Discharge Current	Weight (g)	Dimension (mm)		Cost (₱)
						Height	Diameter	
PKCell 18650	Lithium-ion	3.7V	3350	10A	48.27	65.2	18.5	185



Only two battery cells to supply 3 rail voltage for electrical components.

Information

1. The PKCell 18650 battery supplies 3.7 volts. Single battery has a capacity of 3.7V and 3350 mAh. The battery can generate until 10A continuously.
2. The battery supplies the system until more than 4 hours.
3. One battery can power all electronic components in the CanSat.



9.6 CanSat Power Budget

Component	Quantity	Voltage (V)	Current (mA)	Duty Cycles in 1 Hour (%)	Source	Power Consumption (Wh)
UBLOX – Neo M8N	1	3.3	67	100	Datasheet	0.2211
Buzzer 95 dB	1	3.3	80	100	Datasheet	0.264
MPL3115A2	1	3.3	80	100	Estimated	0.012
ESP32 OLED V2.0 TTGO	1	3.6	60	100	Estimated	0.398
SD Card Module	1	3.3	20	50	Datasheet	0.0033
MPU6050	1	3.3	11	100	Datasheet	0.073
ESP32 CAM	2	5	140	100	Datasheet	0.047
HC12 (Transmit)	1	3.3	215	100	Datasheet	0.213
HC12 (Receive)	1	3.3	26	100	Datasheet	0.0858
MAX4466 Amplifier	1	3.3	50	100	Datasheet	0.165
Voltage Divider	1	3.6	0.878	100	Measured	0.0031608
Total						1.5323608
Consumption for 4 Hours						6.1294432



Selected CanSat Power: PKCell 18650

Available Power = Voltage x Capacity
= 3.7 x 3.35 Ah
= 12.395 Wh

Power Consumption (Wh) = 6.1294432

Margins = Available power – Power Consumption
= 12.395 Wh – 6.1294432 Wh
= 6.266 Wh

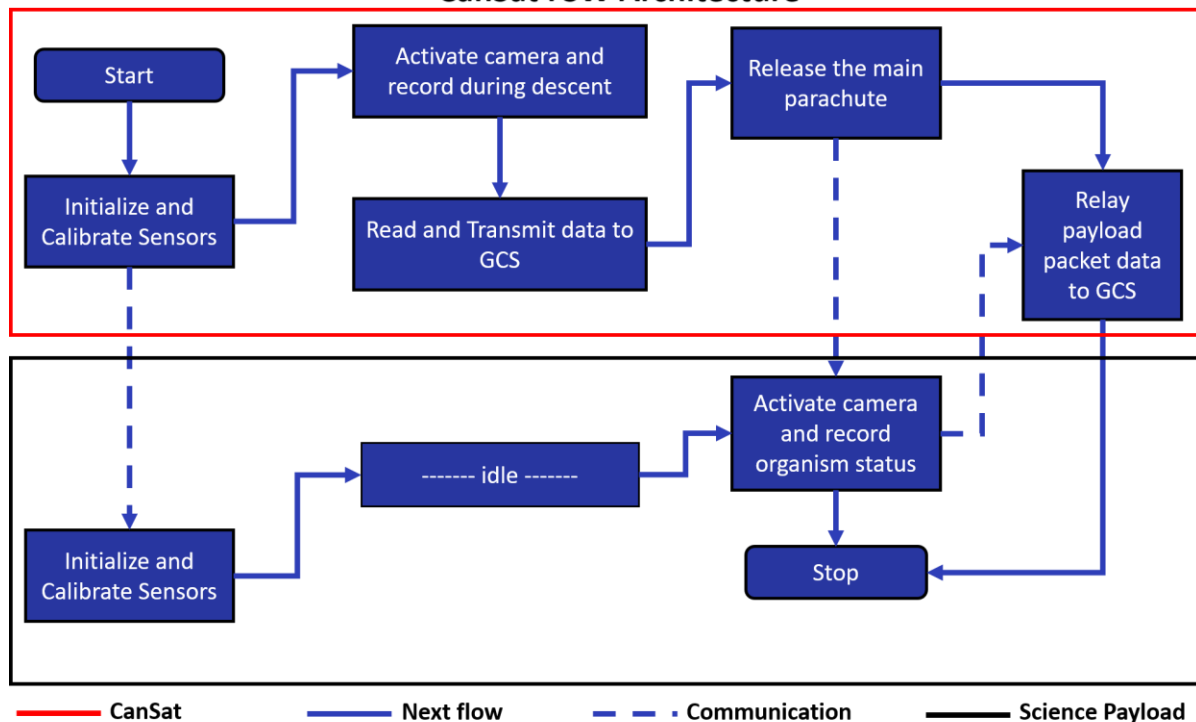


10 Flight Software (FSW) Design

10.1 FSW Overview

CanSat FSW Tasks	
- CanSat will collect data from sensors during ascent until it lands and transmits the data to GCS. Since the launch of the CanSat video camera will record the status of the payload organism. The EEPROM and SD Card are used to back up the data and store the video on the computer. - A video camera will be added to the sCanSat pointed 45 degrees downward in the south direction to assure the terrain is in the video.	
Programming language	Development Environment
C/C++	Arduino IDE

CanSat FSW Architecture

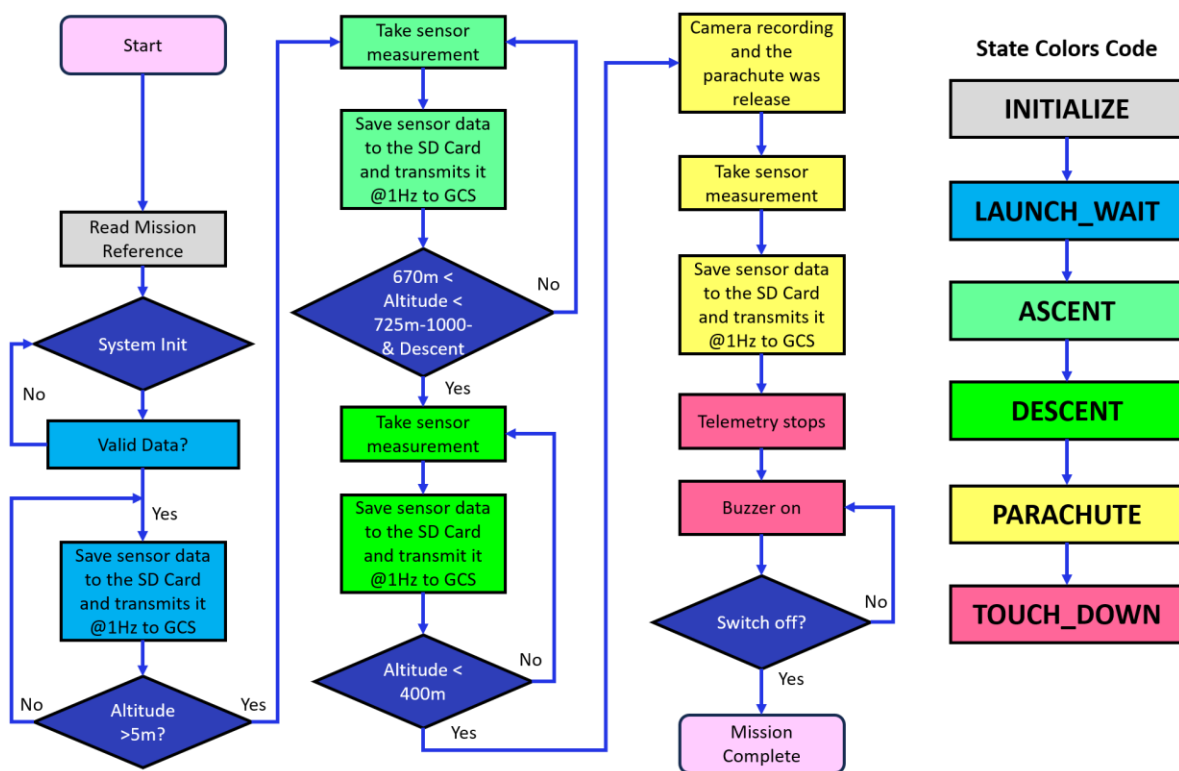




CanSat FSW Task

1. The CanSat will set the new mission reference from GCS and save calibration data to the EEPROM.
2. The CanSat FSW mode will read from EEPROM on the EPS32 OLED V2.0 TTGO, the default mode is Flight mode.
3. The CanSat will be collecting data from the sensors and science payload then save it to the SD CARD, the CanSat data will be transmitted @1Hz rate to GCS via HC-12.
4. The buzzer will be activated when the state is TOUCH_DOWN, and all telemetry transmission will be stopped. The buzzer will beep continuously until the switch is off.
8. The mission will be completed.

10.2 CanSat FSW State Diagram



Major Decision Points in The Logic

Altitude will be the major decision parameter among other parameter used as consideration.

Data Storage	Sampling of Sensor	Communications
1. Video and backup telemetry data will be stored on the SD Card. 2. EEPROM is used for recovery after reset.	The data sensor will be sampled at 1 Hz (1000ms).	Science payload data collection transmits to GCS. All commands will be included in CDH.



10.3 Software Development Plan

Prototyping and Prototyping Environments		
Subject	Prototyping Environment	Prototyping Procedure
TTGO T-Display ESP32	Arduino IDE	Programming and debugging are done in Arduino IDE and the data will be monitored in the serial monitor.
Sensors	Breadboard and PCB	Each sensor is tested on the breadboard separately.

Software Subsystem Development Sequence	
Subsystem	Development Sequence
Sensors	Sensor trade and selection: select the best sensors for our application. Individual sensor programming: program container sensor and program payload sensor with Arduino IDE.
State Mechanism	Integrate all sensors and test it in state mechanism.
HC-12 Telemetry	Testing GCS and CanSat communication: Configure and test all CanSat sensors data that will be transmitted to GCS.
Integrate all	Integrate all software subsystem to ensure all system works well.
Development Team	
1. Cyrill John Cuña 2. Jann Rovic Cueto	

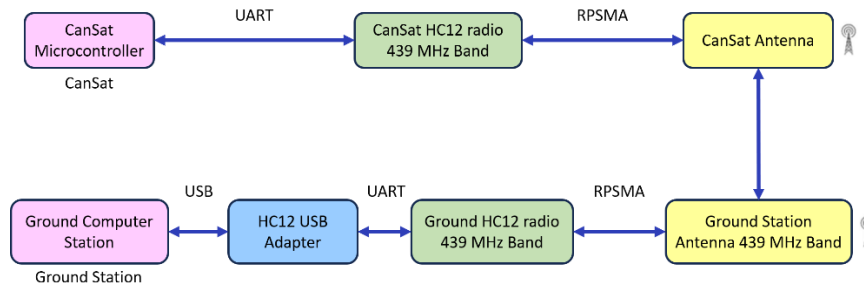
Test Methodology	
1. Necessary software is installed to Arduino IDE and to help the software development. 2. Telemetry software tests are simulated using to Arduino IDE terminal. 3. Sensors and hardware were tested separately. 4. Test the states mechanism for CanSat and scientific payload. 5. Test the system recovery for CanSat.	6. Test the telemetry data and communication commands using hardware. 7. Test the flight mode software using GCS. 8. Check whether the FSW meets the competition requirements. 9. Test integrated sensors and hardware according to mission.
CanSat Development Update	
1. Sensors were calibrated and tested. 2. CanSat flight software states has been tested. 3. CanSat is able to receive commands from GCS properly. 4. CanSat data recovery was tested? 5.CanSat software development has been made based on mission guide.	



11 Ground Control System (GCS) Design

11.1 GCS Overview

We will configure our own Ground Control System, consisting of the following components:



The whole GCS setup is powered from battery and easy to carry.

Specifications	
Operation Time	GCS will be able to operate for at least 4 hours with no external power supply.
Overheating Mitigation	We will protect the computer from sunlight with an umbrella.
Screen Reflection Mitigation	The umbrella will be covered with a blanket. The computer will operate in light mode (black text on white background) with size of 12 points. The plots will be made using bold colors on light background.
Auto update Mitigation	An Asus Ryzen 7 gaming laptop is going to be used with auto updates turned off, and Windows does not enforce auto updates when they are disabled.



11.2 GCS Antenna

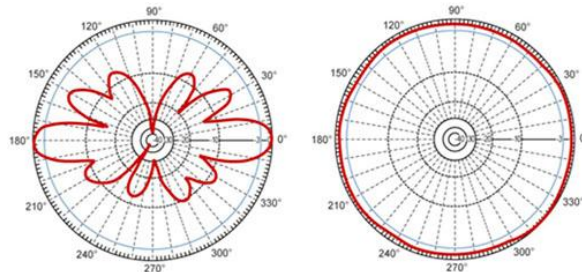
The dipole antenna offers excellent signal reception coverage. Furthermore, this particular antenna model underwent testing in real-world scenarios during past CanSat competitions.



473 MHz omni-directional Rubber Duck Antenna



Model	Type	Frequency range (MHz)	Gain (dBi)	Size (mm)	Direction	Range (km)
Quarter -wave antenna 473	Dipole	433-473	6	19.5x1.3cm	Omnidirectional	1- 3.5



Vertical

Horizontal

Radiation patterns comparison

11.3 Antenna Link Budget

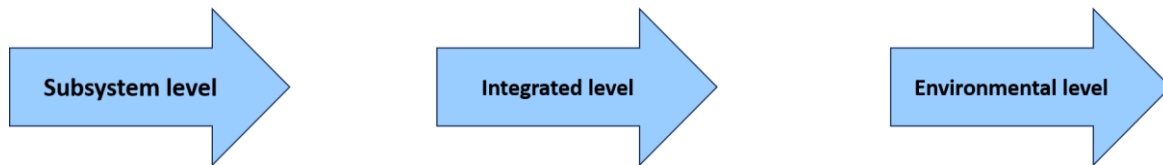
Link Budget Calculation	Information
$P_r = P_T + G_T + G_R - FSPL$ $P_r = 20 + 2 + 2 - 85.17$ $P_r = -61.17dBm$	P_T = Transmitted Power Output (dBm) G_T = Transmitter Antenna Gain (dBi) P_r = Receiver Sensitivity (dBm)
Free Space Path Loss Calculation	Information
$FSPL = 20\log_{10}(d) + 20\log(f) + 20\log_{10}\left(\frac{4\pi}{c}\right)$ $FSPL = 20\log_{10}(1000) + 20\log(439 \times 10^6)$ $+ 20\log_{10}\left(\frac{4\pi}{3 \times 10^8}\right)$ $FSPL = 85.291dB$	$FSPL$ = Free - Space Path Loss (dB) d = distance between antennas (km) f = Frequency (MHz)
Distance = 1km (predicted)	Therefore, the received signal is still strong

12 CanSat Integration and Test

Our CanSat will be tested by four level testing		
Subsystem level	All the subsystems will be tested separately	Sensors, CDH, EPS, Radio Communications, FSW, Mechanical, Descent Control
Integrated level	Subsystems will be integrated and tested together	Descent Testing, Communications, Mechanisms, Deployment



Environmental level	Whole device will be subjected to extreme conditions and final mechanical check will be performed.	Drop test, Thermal test, Vibration test, Fit Check, Vacuum test
----------------------------	--	---



1. Firstly, the payload, CanSat body and mechanism will be built and initially tested. Independently, the electronics circuits will be soldered, programmed and debugged.
2. When all of the components are working properly, they will be put together and tested as a whole device to ensure that all mechanism and electronics work together properly.
3. Testing of all algorithm will ensure that it was implemented properly. Finally, the environmental tests will be conducted to make sure that our CanSat can work within specified environment.

12. 1 Subsystem Level Testing Plan

Subsystem	Test Case	Acceptance Criteria	Test Result
FSW	Microcontroller is correctly initialized	Debugger is connecting	Tested successfully
	Debug logs can be sent via serial interface	Serial interface transmits data	Tested successfully
	After initialization FSM jumps to state selector	Debug log indicates FSM entered state selector	Tested successfully
	FSW can progress through mission states	Each state sends signed debug log	Tested successfully
	FSW can survive power cuts	FSM starts with last state; sensor calibration data is remembered. Date and time are correct	Tested successfully
Mechanical	Payload must maintain its structural integrity after experiencing shock and vibrations	No cracks or loose bolts must be detected after a shock test, vibration test and simulated landing	Tested successfully
	The biological membrane compartment can protect the microorganism from impact	The microorganism glass slide and glass cover aren't cracked after a simulated landing	Upcoming
	CanSat parachute opens after ejecting in rocket	Parachute opens smooth and parachute cords are not tangled	TBD



	Main electronic board is isolated from vibrations	Electronics does not reset; all communications interfaces are working	Tested successfully
Sensors	Sensors and devices are initialized, communication is possible	Correct sensor/device data can be read via debug logs	Tested successfully
	Altitude can be properly calculated	Altitude calculated from test values corresponds to expected altitude	Tested successfully
	Temperature measuring	Temperature is correctly measured	Tested successfully
	Pressure measuring	Pressure is correctly measured	Tested successfully
	Voltage measuring	Voltage is correctly measured	Tested successfully
	GPS data is available	GPS data can be read by FSW	Tested successfully
	GPS frame is correctly decoded	GPS time, latitude, longitude, altitude and sats are available	Tested successfully
	Sensors can be calibrated and return zero values when stationary on ground	Values read from sensors are zeroed	Tested successfully
CDH	File system is properly initialized on SD Card	File system initialization success, SD Card can be read by a windows laptop	Tested successfully
	SD Card is ready for writing data	File system returns success when saving files	Tested successfully
	SD Card is accessible; data can be read	File system allows fore reading files, data in files are correct	Tested successfully
	Data saved on SD Card survives resets and power downs	Files are consistently available	Tested successfully
	Real-time Clock measures time	Times of CanSat and GS are in sync during the tests	Tested successfully
EPS	Pressure sensor is powered	Proper voltage on necessary pins is measured	Tested successfully



	Temperature sensor is powered	Proper voltage on necessary pins is measured	Tested successfully
	GPS sensor is powered	Proper voltage on necessary pins is measured	Tested successfully
	Voltage sensor is powered	Proper voltage on necessary pins is measured	Tested successfully
	Camera is powered	Proper voltage on necessary pins is measured	Tested successfully
	Processor is powered	Proper voltage on necessary pins is measured	Tested successfully
	MicroSD card is powered	Proper voltage on necessary pins is measured	Tested successfully
	PCB doesn't have shortcuts	PCB has no shorts to ground or power lines	Tested successfully
	PCB is properly soldered	Joints are properly soldered, there are no cold joints	Tested successfully
	Battery withstands high current	Battery works correctly under high current	Tested successfully
	CanSat can operate for 4 hours	CanSat does not turn off when left in IDLE state for 4 hours	Tested successfully
	Voltage regulators and DC converters work properly	Proper voltage on all power lines under load	Tested successfully
	IMU is powered	Proper readings from the sensor given the positioning	Tested successfully
	HC12 is powered	Proper voltage on necessary pins is measured	Tested successfully
Descent Control	Parachute flight maintain correct descent velocity	CanSat descends at rate between 5 m/s to 12 m/s	Tested successfully
Radio Communication	HC12 radio can receive data	Data sent to HC12 can be read via debug logs	Tested successfully
	HC12 radio can transmit data	Data sent from FSW can be read on Laptop with HC12	Tested successfully



	HC12 radios channel/frequency is set to their team number	Frequency is set to 439 MHz and channel to 015	Tested successfully
	Ground level radio range testing	Radio range is sufficient for the flight altitude	Tested successfully
	GCS can connect with the CanSat	The connection is reached and stable	Tested successfully
	GCS can transmit and receive data	Data successfully transmitted and received	Tested successfully
	GCS plotting data	Data is plotted in real time	Tested successfully

12.1 Integrated Level Functional Test Plan

Descent Testing

1. This test is to verify that CanSat descends at a speed corresponding to mission guide.
2. We tied a dummy CanSat to a drone with mass of 340 g to verify its descent rate.

Documentations

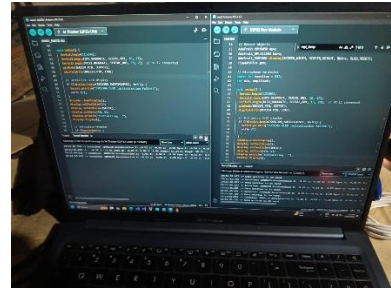
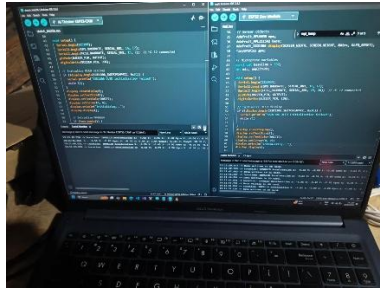


Communications

1. This test is to verify that the communication system worked properly.
2. We will run the flight software to run the HC12 communication at 1Hz for the CanSat. The data shall appear in GCS monitor.
3. Ensure data received is in correct format to mission guide.



Documentations



Deployment

1. Deploying CanSat and science payload in various altitude.
2. Parachute deployment will be tested in various altitude.
3. Ensure there is no sharp edges and obstacle that prevents CanSat from deploying.

Software Testing

1. This test is designed to verify if Ground Station shall have the capability to read a .csv file of barometric pressure data that simulates the mission profile and receive the values from the CanSat at a rate of one per second (1Hz).
2. We will test it by preparing barometric data in .csv file to ensure GCS is able to read .csv file.
3. Test if GCS is able to receive the pressure value at one second interval (1Hz) from flight software.

12.2 Environmental Test

Drop Test

This test is designed to verify that the CanSat parachute and attachment point will survive the deployment from the rocket payload section which can be very violent.

1. Power on CanSat.
2. Verify telemetry is being received.
3. Raise CanSat by a 61 cm non-stretching cord.
4. Release CanSat.
5. Verify the CanSat did not lose power. Inspect for any damage, or detached parts and verify telemetry is still being received.



Thermal Test

This test is to verify the CanSat can operate in a hot environment. We will be using a thermal chamber and hot air gun as heat source.

1. Place the CanSat into a thermal chamber and Hot Air Gun as heat source.
2. Turn on the CanSat.
3. Close and seal the thermal chamber.
4. Turn on the heat source.
5. Turn off the heat source when the internal temperature reaches 60 °C and turn on the heat source when the temperature drops to 55 °C for two hours.
6. Perform visual inspection and any functional tests to verify the CanSat survived the thermal exposure.



Reference Test Fixture

Vibration Test

This test is designed to verify the mounting integrity of all components, mounting connections, structural integrity, and battery connections. We will be using orbital sander/drill for vibration test.

1. Power on the CanSat and verify accelerometer data is being collected.
2. Plaster CanSat above the sander and power up the orbital sander.
3. Once the sander is up to full speed, wait 5 seconds.
4. Power down the sander to a full stop.
5. Repeat it four more times.
6. Inspect the CanSat for damage and functionality. Verify accelerometer data is still being collected.



Reference Test Fixture



Vacuum Test	
This test is designed to verify deployment operation of the payload. 1. Suspend the fully configured and powered CanSat in the vacuum chamber. 2. Turn on the vacuum to start pulling a vacuum. 3. Monitor the telemetry and stop the vacuum when the peak altitude has been reached. 4. Let the air enter the vacuum chamber slowly and monitor the operation of the CanSat. 5. Collect and save telemetry.	 Reference Test Fixture

Fit Check
This test is designed to verify if the CanSat is able to fit in rocket. We use micrometer calipers to measure the dimensions of CanSat to ensure it is able to fit in rocket.

13 Test Procedures Descriptions

Test Proc	Test Description	RN	Pass/Fail Criteria	Status
1	Air pressure sensor test		Altitude from data is similar with actual altitude.	Pass
2	Air temperature sensor test		Temperature data is similar with actual temperature.	Pass
3	GPS sensor test		Showing correct time and location.	Pass
4	Camera test		Result of video capturing minimum resolution of 640x480 and 30 fps.	Pass
5	Rotational sensor test		RPM data is similar with tachometer.	Pass
6	Power consumption test		Actual power consumption similar with calculation result measured using Avometer.	Pass
7	HC12 communication test		Communication data sent and receive by GCS from 750m-100m distance.	Pass
8	Data transmission at 1Hz test		The data must be in correct order at 1 Hz rate.	Pass
9	FSW state test		State changes along as altitude increases and descends that calculated from pressure.	Pass



10	Testing of data recovery algorithm		The data (altitude reference, packet count, states, coordinate, etc.) recovered to EEPROM and continuing process.	Pass
11	GCS interface test		All telemetry data received are displayed in GCS interface (numerical or graph).	Pass
12	GCS PC battery test		GCS Pc is still working after 4 hours of usage.	Pass
13	GCS data inspection		The data displayed in engineering unit.	Pass
14	GCS communication test		GCS is able to send command to CanSat to start send the telemetry data.	Pass
15	CanSat descent test		CanSat descends at descent rate of 10m/s (± 5 m/s) for the parachute	Pass
16	Payload camera test		Payload camera is clearly records the organism behavior.	Upcoming
17	CanSat deployment test		CanSat doesn't have any sharp edges	Pass
18	Parachute deployment test		Parachute able to deploy from CanSat successfully.	Pass
19	Payload preparation test		Payload organism ensure properly prepared and alive inside the CanSat.	Upcoming
20	CanSat maintain orientation test		CanSat able to maintain orientation of a video camera pointing in the south direction and video camera shall be pointed 45 degrees downward to assure terrain is in the video.	Upcoming
21	Drop test		CanSat survived a drop from 2 feet attached to cord. No parts falling off, no separation.	Pass
22	Thermal test		No CanSat materials warp, weaken, change characteristics, or fail to function.	Upcoming
23	Vibration test		All mounting integrity of all components; mounting connections, structural integrity, and battery connections survived.	Upcoming
24	Vacuum test		FSW is able to detect the altitude from pressure changes inside the vacuum chamber.	Upcoming
25	Fit Check		Total Mass of CanSat 350 g +/- 10 grams and dimension of CanSat under 115 mm x 66mm.	Pass



14 Mission Operation & Analysis

14.1 Mission Sequence of Events

1. Arrival	• Team Arrival at launch site • Set up GCS and antenna • Score sheet review and checklist
2. Pre-Launch	• Communication inspection • Mechanism inspection • CanSat and Payload Assembly • Ensure the organism payload properly set up • CanSat dimension and weight check
3. Rocket Integration	• Final Inspection before launch • Collect CanSat, turn it on, integrate into rocket and ensure communication with GCS
4. Mission	• Rocket launch with CanSat inside it • Flight monitoring • Display GCS to the judges and collecting telemetry data during mission • Recovery crew preparation
5. Recovery	• Search for CanSat by fluorescent color, audio beacon and GPS • Retrieve data from SD Card of the CanSat.
6. Data Analysis	• GCS data analysis and acquisition • Deliver SD card and telemetry data to judges for scoring • Launch day team evaluation • PFR preparation

Roles	Member Name
Mission Control Officer - Responsible for informing the Flight Coordinator when the team and their CanSat is ready to be launched.	1. Frank Fitzgerald Batin
Ground Station Crew - Responsible for monitoring the ground station for telemetry reception and issuing commands to the CanSat.	1. Jann Rovic Cueto 2. Cyrill John Cuña 3. Roland Dayrit
Recovery Crew - Responsible for tracking the CanSat and going out into the field for recovery and interacting with the field judges.	1. Jann Rovic Cueto 2. John Allen Tandoc
CanSat Crew - Responsible for preparing the CanSat, integrating it into the rocket, and verifying its status.	1. Roland Dayrit 2. John Allen Tandoc 3. Cyrill John Cuña



Antenna construction and ground system setup	CanSat Assembly
1. Prepare GCS equipment. 2. Connect HC12 to laptop by adapter and connect Antenna to HC12 by SMA connector. 3. Turn on laptop and disable windows auto update. 4. Perform communication test to ensure all data retrieved. 5. Antenna held by team member.	1. Assemble electrical component. 2. Mount each component and stow the science payload inside CanSat. 3. Fold parachute. 4. Turn on CanSat.
CanSat Test	Delivery of Telemetry Data File
1. CanSat and mass check. 2. Drop Test. 3. Parachute mechanism test. 4. Antenna and communication test. 5. Sensor calibration and electronics component check. 6. CanSat final check.	1. GCS receive telemetry data from HC12. 2. GCS save received telemetry data in .csv file format. 3. Retrieve .csv from file manager into PC and move it into USB flash drive. 4. GCS operator delivers .csv file using USB drive.

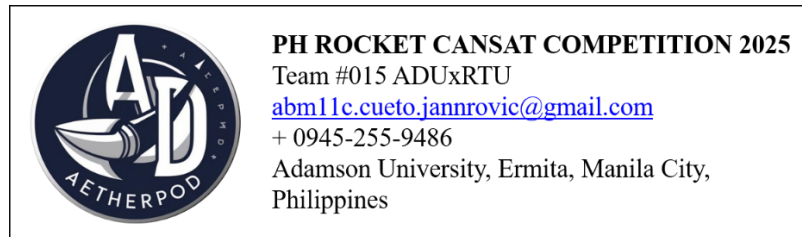
14.2 Field Safety Rules Compliance

Mission Operations Manual	Description
Ground Station Configuration	1. Laptop and antenna preparation 2. Communication check and testing
CanSat Preparation	1. CanSat assembly 2. Sensor calibration 3. Mass and size checking 4. Calibration system by command from GCS
CanSat Integration	1. Fit check 2. Drop test 3. Battery verification
Launch Preparation	1. Final communication check 2. Check CanSat stowed state inside rocket
Launch Procedure	1. Pre-launch checks 2. Verify with the ground station that the ground station is ready for launch
Development Status	
1. There will be a Mission Operations Manual will be assembled to a ring binder. It consists of five section and each section will start on its own page according on mission guide book of CanSat. 2. Each member will familiarize themselves with the manual before launch day. Safety of each person on launch day is crucial. 3. Mission Operations Manual will be provided before the event schedule. It will be updated as we encounter changes before launch day.	



14.3 CanSat Location and Recovery

CanSat Recovery
1. Recovery crew will maintain visual contact with the CanSat and payload to aid recovery. 2. CanSat and payload will be blue/yellow colored to ease recovery process. 3. We will be providing team information on the outer structure of CanSat. 4. Container has a buzzer that will continuously buzzing when it landed. 5. We also track the CanSat through GPS.



We will place this sticker to our CanSat body

15 Requirement Compliance

We have prepared and designed CanSat by analyzing and identifying the CanSat Competition Mission Guide 2025. The system will be tested according to CanSat Integration and Test section.

- We have complied thirty-eight requirements based on CanSat Competition Mission Guide 2025.
- There are two partially complied requirements that require further tests. Improvement will be needed to comply our system to get fully comply and it's discussed at CDR phase.
- There are not any requirements that doesn't comply to our design.

- = Complied
- = Partial
- = No Complied

RT	Requirements	Compliance
RT#1	All the components of the CanSat must fit a standard soft drink can (115 mm height and 66 mm diameter), with the exception of the parachute.	Complied
RT#2	Radio antennas and GPS antennas can be mounted externally on the top or bottom of the can, depending on the design, but not on the sides.	Complied
RT#3	The antennas, transducers and other elements of the CanSat cannot extend beyond the cans diameter until it has left the launch vehicle.	Complied



RT#4	The mass of the CanSat must be between a minimum of 300 grams and a maximum of 350 grams +/- 10 grams. CanSat that are lighter must take additional ballast with them to reach the 300 grams minimum mass limit required.	Complied
RT#5	Explosives, detonators, pyrotechnics, and inflammable or dangerous materials are strictly forbidden. All materials used must be safe for the personnel, the equipment, and the environment.	Complied
RT#6	The CanSat must be powered by a battery and/or solar panels. It must be possible for the systems to remain switched on for four continuous hours.	Complied
RT#7	The battery must be easily accessible in case it has to be replaced/recharged.	Partial
RT#8	The CanSat must have an easily accessible master power switch.	Complied
RT#9	Global Positioning System or GPS and a Beeper is required in the CanSat.	Complied
RT#10	The CanSat should have a recovery system, such as a parachute, capable of being reused after launch. It is recommended to use bright colored fabric, which will facilitate recovery of the CanSat after landing.	Complied
RT#11	The parachute connection must be able to withstand up to 50 N of force. The strength of the parachute must be tested to ensure that the system will operate nominally.	Complied
RT#12	A descent rate between 8 and 11 m/s is recommended for recovery reasons. However, the CanSat's descent speed must not be lower than 5 m/s or higher than 12 m/s for safety reasons.	Complied
RT#13	The CanSat must be able to withstand an acceleration of up to 20g.	Complied
RT#14	A Sd card module for data logger that saves data is required in the CanSat. And the cost of the CanSat must not exceed 8, 000 pesos.	Complied
RT#15	All electronics shall be hard mounted using proper mounts such as standoffs, screws, or high-performance adhesives.	Complied
RT#16	All mechanisms shall be capable of maintaining their configuration or states under all forces.	Complied
RT#17	Both the CanSat and payload shall be labeled with team contact information including email address.	Complied
RT#18	The CanSat shall include a pressure sensor to measure altitude.	Complied
RT#19	The CanSat shall measure its battery voltage.	Complied
RT#20	The CanSat shall transmit its telemetry once per second (1 Hz) in the formats described in the Telemetry Requirements section.	Complied
RT#21	The science payload shall include a video camera pointing directly in the microorganism sample.	Complied



RT#22	The CanSat shall include a video camera pointing 45 degrees down from the CanSat NADIR direction.	Complied
RT#23	The CanSat payload shall maintain orientation so the camera always faces south within +/- 20 degrees.	Complied
RT#24	The ground station shall command the CanSat to start transmitting telemetry prior to launch.	Complied
RT#25	The ground station shall generate csv files of all sensor data as specified in the Telemetry Requirement Section.	Complied
RT#26	Telemetry shall include mission time with one second or better resolution. Mission time shall be maintained in the event of processor reset during the launch and mission.	Complied
RT#27	Configuration states such as if commanded to transmit telemetry shall be maintained in the event of a processor reset during launch and mission.	Complied
RT#28	Each team shall develop their own ground station.	Complied
RT#29	All telemetry shall be displayed in real time during descent on the ground station.	Complied
RT#30	All telemetry shall be displayed in engineering units (meters, meters/sec, Celsius, etc.)	Complied
RT#31	Teams shall plot each telemetry data field in real time during flight (optional).	Partial
RT#32	The ground station shall include one laptop computer with a minimum of four hours of battery operation, HC12 radio.	Complied
RT#33	All video cameras shall be in color, have a resolution of at least 640x480 and record at a minimum of 30 frames a second.	Complied
RT#34	The ground station must be portable so the team can be positioned at the ground station operation site along the flight line. AC power will not be available at the ground station operation site.	Complied
RT#35	The science payload shall include a video camera pointing 45 degrees down from the payload NADIR direction.	Complied
RT#36	The CanSat must operate during the environmental tests.	Complied
RT#37	The flight software shall maintain a count packet transmitted which shall increment with each packet transmission throughout the mission. The value shall be maintained through processor resets.	Complied
RT#38	The CanSat shall maintain mission time throughout the whole mission even with processor resets or momentary power loss.	Complied
RT#39	The CanSat shall have its time set to UTC time to within one second.	Complied
RT#40	The flight software shall use the radio uplink pressure values in place of the pressure sensor for determining the CanSat altitude.	Complied



16 Management

16.1 Mechanical Procurement

Components	Quantity	Order	Receive	Status
PETG Filament	1	02-24-2025	02-24-2025	Received
Black Nylon Umbrella	1	n/a	n/a	In-stock
Screws	10	n/a	n/a	In-stock
Fish string	2	n/a	n/a	In-stock
Acrylic A3 2mm	1	02-24-2025	02-24-2025	Received
Macroscope and Acrylic A3 2mm	1	02-24-2025	02-24-2025	Received

16.2 Electronics Procurement

Components	Quantity	Order	Receive	Status
UBLOX-Neo M8N	1	02-10-2025	02-10-2025	Received
Buzzer	1	02-10-2025	02-10-2025	Received
ESP32 OLED V2.0 TTGO	1	02-10-2025	02-10-2025	Received
SD Card Module	1	02-10-2025	02-10-2025	Received
MPU6050	1	02-10-2025	02-10-2025	Received
ESP32 CAM	2	02-10-2025	02-10-2025	Received
HC12 439Mhz - 1000 m (Transmit)	1	02-10-2025	02-10-2025	Received
MAX4466 Amplifier	1	02-10-2025	02-10-2025	Received
PKCell 18650 Battery	2	02-10-2025	02-10-2025	Received
SanDisk Ultra 16 GB SD Card	3	02-10-2025	02-10-2025	Received
PCB	1	02-10-2025	02-10-2025	Received
MPL3115A2	1	02-10-2025	02-10-2025	Received
Thermistor	3	02-10-2025	02-10-2025	Received
Battery Holder 18650 (2 Slot)	1	02-10-2025	02-10-2025	Received

16.3 Ground Station Procurement

Components	Quantity	Order	Receive	Status
Antenna	1	02-24-2025	02-24-2025	Received



HC12 439Mhz - 1000 m (Receive)	1	02-10-2025	02-10-2025	Received
SMA connector and 30 cm RF Cable	1	02-24-2025	02-24-2025	Received
CP2102 USB 2.0	1	02-24-2025	02-24-2025	Received

16.4 CanSat Budget – Mechanics Components

Components	Quantity	Unit Cost (₱)	Total Cost (₱)	Considerations
PETG Filament	1	749	749	Actual
Macroscopic and Acrylic A3 2mm	1	569	569	Actual
Total Cost Mechanics Component (₱)			1,318	

16.5 CanSat Budget – Electrical & Electronics Components

Components	Quantity	Unit Cost (₱)	Total Cost (₱)	Considerations
UBLOX-Neo M8N	1	600	600	Actual
Buzzer	1	10	10	Actual
ESP32 OLED V2.0 TTGO	1	1099	1099	Actual
SD Card Module	1	65	65	Actual
MPU6050	1	85	85	Actual
ESP32 CAM	2	499	998	Actual
HC12 439Mhz - 1000 m (Transmitter)	1	385	385	Actual
MAX4466 Amplifier	1	529	529	Actual
PKCell 18650 Battery	2	185	370	Actual
SanDisk Ultra 16 GB SD Card	3	180	540	Actual
PCB	2	35	70	Actual
MPL3115A2	1	645	645	Actual
Thermistor	3	14.3	42.9	Actual
Resistor (pack)	1	65	65	Actual
Battery Holder 18650 (2 Slot)	1	300	300	Actual
Total Cost Electrical and Electronics Components (₱)			5,803.9	



16.6 CanSat Budget – Ground Station Components

Components	Quantity	Unit Cost (₱)	Total Cost (₱)	Considerations
Antenna	1	100	100	Actual
HC12 439Mhz - 1000 m (Receiver)	1	385	385	Actual
SMA connector and 30 cm RF Cable	1	200	200	Actual
CP2102 USB 2.0	1	139	139	Actual
Total Cost Ground Station Component (₱)			824	

16.7 CanSat Budget – Total Hardware Cost

Hardware Parts	Total (₱)
Total Cost Mechanics Components	1318
Total Cost Electronics components	5803.9
Total Cost Ground Station Components	824
Total Hardware Budget (₱)	7,945.9

CanSat Requirement Cost – Hardware Budget = **Margin**

$$₱ 8000 - ₱ 7945.9 = ₱ 54.1$$

16.8 CanSat Budget – Other Cost

Components	Quantity	Unit Cost (₱)	Total Cost (₱)	Considerations
Philippine Airlines Flight (MNL - CEBU)	5	2795	13975	Estimated
Uniform	5	455		Actual
Car Rental	1	2500	3500	Estimated
Food	6	2000	2000	Estimated
Total Other Cost (₱)			19,475	

16.9 CanSat Budget – Income

Sources of Income	Total Cost (₱)
Adamson University	10,000
Indiana Aerospace University	20,000
Total Income (₱)	30,000



16.10 Detailed Program Schedule

Task	Assign	Start	End	Days	Status	Dec	Jan	Feb	Mar
Mechanical									
Mission Guide Study	MC	01-Dec-24	28-Dec-24	28	100%				
CanSat Design	MC	29-Dec-24	03-Feb-25	36	100%				
Material Trade	MC	13-Dec-24	19-Jan-25	37	100%				
Procurement of Materials	MC	10-Dec-24	20-Jan-25	41	Ongoing (80%)				
Mass Budget Calculate	MC	13-Feb-25	16-Feb-25	3	Ongoing (90%)				
Prototype Manufacturing	MC	04-Feb-25	12-Feb-25	8	100%				
Prototype Testing	MC	18-Feb-25	21-Feb-25	3	100%				
System Integrating	MC	21-Feb-25	01-Mar-25	8	100%				
System Testing	MC	21-Feb-25	01-Mar-25	8	Ongoing (50%)				
System Improvement	MC	21-Feb-25	05-Mar-25	12	Ongoing (50%)				
Hardware									
Mission Guide Study	HW	01-Dec-24	29-Dec-24	28	100%				
Component Trade	HW	16-Dec-24	2-Jan-25	17	100%				
Component Testing	HW	22-Dec-24	22-Jan-25	31	100%				
Flight Algorithm	HW	22-Jan-25	07-Feb-25	16	100%				
Procurement of Components	MC	10-Dec-24	20-Jan-25	41	Ongoing (80%)				
CanSat PCB Design	HW	25-Jan-25	15-Feb-25	21	Ongoing (90%)				
HC12 Communication Test	HW	10-Feb-25	16-Feb-25	6	100%				
Electrical Prototype Test	HW	11-Feb-25	18-Feb-25	7	100%				
System Integrating	HW	10-Feb-25	2-Mar-25	20	Ongoing (70%)				
System Testing	HW	10-Feb-25	2-Mar-25	20	Ongoing (50%)				
System Improvement	HW	11-Feb-25	5-Mar-25	22	Ongoing (50%)				
Camera Tracking Testing	HW	23-Jan-25	07-Feb-25	15	Ongoing (70%)				
Software									
Mission Guide Study	SW	01-Dec-24	29-Dec-24	28	100%				
GCS Design	SW	9-Jan-25	11-Feb-25	33	100%				
Antenna Trade	SW	16-Dec-24	22-Jan-25	37	100%				
Antenna Development	SW	18-Jan-25	05-Feb-25	18	100%				
Antenna Range Test	SW	30-Jan-25	12-Feb-25	13	Ongoing (70%)				
System Integrating	SW	10-Feb-25	2-Mar-25	20	Ongoing (80%)				
System Testing	SW	10-Feb-25	2-Mar-25	20	Ongoing (50%)				
System Improvement	SW	11-Feb-25	5-Mar-25	22	Ongoing (50%)				
Others									
CDR Submission	SW	01-Dec-24	28-Feb-25	89	Ongoing (70%)				
Travel Planning	SW	25-Feb-25	12-Mar-25	15	Ongoing (50%)				
Competition	SW	10-Mar-25	14-Mar-25	4	Ongoing				

17 Shipping and Transportation

Transportation	
Person	Adamson University – Manila: LRT 3 Train NAIA Terminal 3 – Pasay: Grab Car IAU - Cebu City: Philippine Airlines
Shipping	
CanSat and Tools	Adamson University – Manila: LRT 3 Train NAIA Terminal 3 – Pasay: Grab Car IAU - Cebu City: Philippine Airlines

18 Conclusion

Division	Major Accomplishments	Major Unfinished Work	Testing to Complete
Hardware	1. All sensors has been tested.	1. Integration with software.	1. Payload camera orientation test.



	2. HC12 communication test completed. 3. Flight software test completed and currently being tested into full system.		
Mechanical	1. CanSat is finished and is being used to fit the actual payload. 2. Payload culture testing completed. 3. Several environmental tests have been completed.	1. Design improvement of CanSat based on testing results and evaluation. 2. Overall system integration.	1. Environmental test.
Software	1. System Design completed. 2. Antenna has been built and tested.	1. Integration with hardware.	-
Administration	1. Sponsorship and partnership contracted. 2. Travel and shipment plans have been established.	Waiting for other sponsors.	-
Brand Team	1. Official social media still active. 2. Social media expansion to LinkedIn and Facebook	1. Developing the social media and website promotion.	-



Our picture with PH STARLAB LOGO OR ADU LOGO or anything designs the team wants

Aetherpod Are Ready to Proceed to The Next Stage of Development

1. Critical Design Phase is completed for mechanical, software, electronic systems.
2. Major mechanisms has been tested and there will be further improvements to assure it runs without a problem.
3. Travel and shipment plans have been establisher for competition day in Medelin Cebu City, Philippines.